Templating

08

Environment specific configurations

```
~$ cat pod_production.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: production
~$ _
```

```
~$ cat pod_production.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: production
~$ _
```


~$ ls -l
-rw-r--r-- pod_development.yaml
-rw-r--r-- pod_production.yaml
-rw-r--r-- pod_staging.yaml
~$ _

```
~$ cat pod_staging.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: staging
~$ _
```

Search and replace

```
~$ cat pod_template.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: %ENV_NAME%
~$ _
```



```
~$ sed 's/%ENV_NAME%/production/g' \
   pod_template.yaml > pod_production.yaml

~$ _
```

Many options

Many options

yq

Many options

yq

Kustomize

Many options

yq

Kustomize

Helm

Many options

yq

Kustomize

Helm

kro

yq

Read

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: prod

~$ yq '.spec.containers[0].image' pod.yaml
```

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: prod

~$ yq '.spec.containers[0].image' pod.yaml
k8s.gcr.io/busybox
```

Read values

Read values

Add/update values

Read values

Add/update values

Generate YAML

Read values

Add/update values

Generate YAML

YAML 2 JSON

Read values

Add/update values

Generate YAML

YAML 2 JSON

Merge

yq

Update

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: prod

~$ yq '.metadata.name = "learnk8s"' pod.yaml
```

```
~$ yq '.metadata.name = "learnk8s"' pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: learnk8s
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: prod

~$ _
```

yq

Merge

```
~$ cat proxy.yaml

apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
  - name: proxy
    image: envoy

~$ _
```

```
~$ cat proxy.yaml

apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
  - name: proxy
    image: envoy

~$ _
```

```
~$ cat app.yaml

apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
  - name: app
    image: busybox

~$ _
```

```
~$ yq '. *+ load("proxy.yaml")' app.yaml
```

```
~$ yq '. *+ load("proxy.yaml")' app.yaml

apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: app
      image: busybox
    - name: proxy
      image: envoy

~$ _
```

Merge from the top

File order matters

Operators

yq multiply

Merge from the top

File order matters

Operators

Merge from the top

File order matters

Operators

yq multiply

Merge from the top

File order matters

Operators

`+`	append arrays	`?`	only merge existing fields
`d`	deeply merge arrays	`n`	only merge new fields

Kustomize

```
~$ cat proxy.yaml

apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
  - name: proxy
    image: envoy

~$ _
```

```
~$ cat app.yaml

apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
  - name: app
    image: busybox

~$ _
```

```
~$ cat kustomization.yaml

apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - app.yaml
patches:
  - path: proxy.yaml

~$ _
```



```
~$ cat kustomization.yaml

apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - app.yaml
patches:
  - path: proxy.yaml

~$ _
```

```
~$ cat kustomization.yaml

apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - app.yaml
patches:
  - path: proxy.yaml

~$ _
```

```
~$ kubectl kustomize .
```

```
~$ kubectl kustomize .

apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: proxy
      image: envoy
    - name: app
      image: busybox

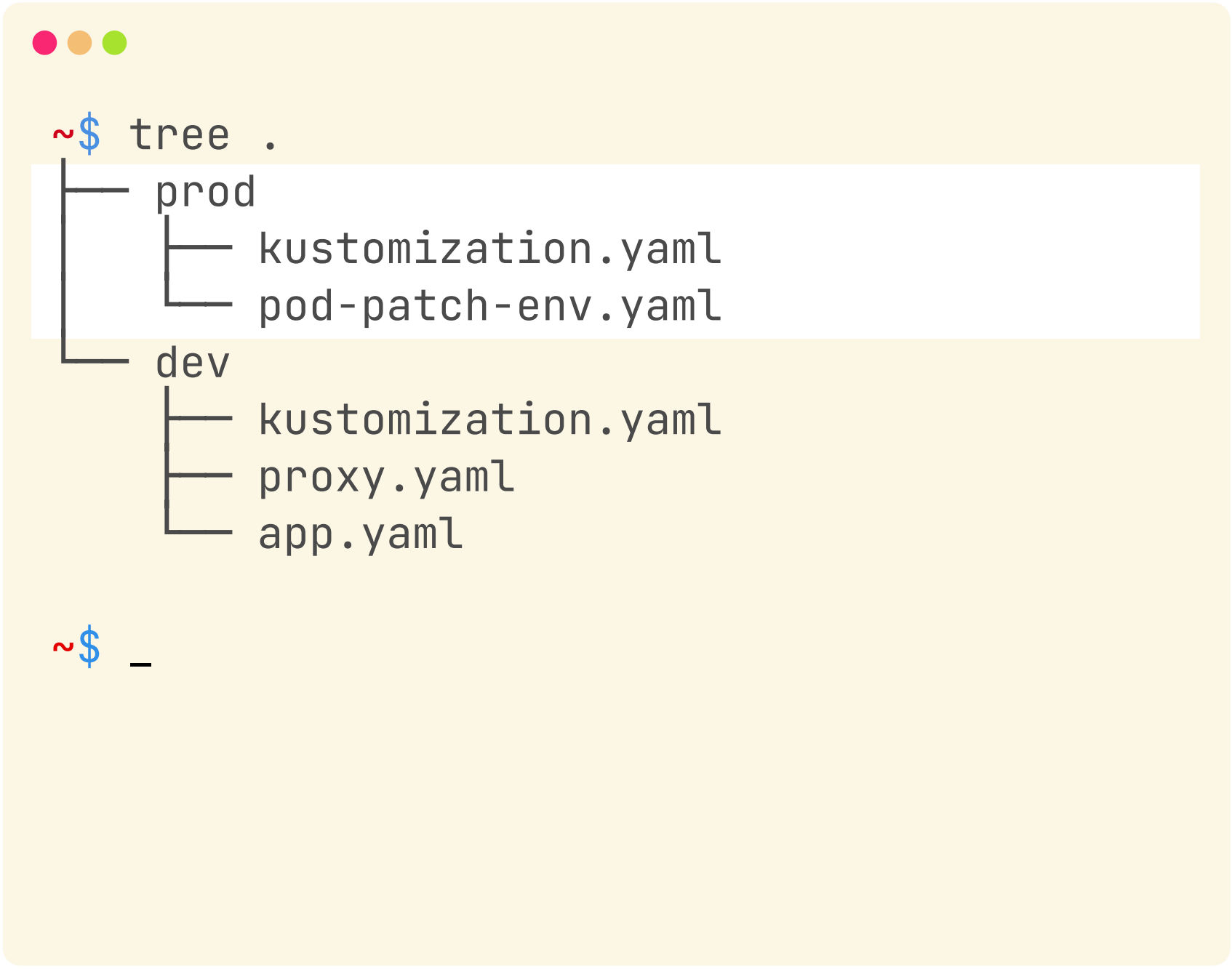
~$ _
```

Kustomize in practice



```
~$ tree .
├── prod
└── dev
    ├── kustomization.yaml
    ├── proxy.yaml
    └── app.yaml

~$ _
```


~$ tree .
prod
kustomization.yaml
pod-patch-env.yaml
dev
kustomization.yaml
proxy.yaml
app.yaml
~$ _



```
~$ cat prod/kustomization.yaml

apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - ../dev
patches:
  - path: pod-patch-env.yaml

~$ _
```

```
~$ cat prod/kustomization.yaml

apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
  - ../dev
patches:
  - path: pod-patch-env.yaml

~$ _
```

```
$ kubectl kustomize prod
```

---

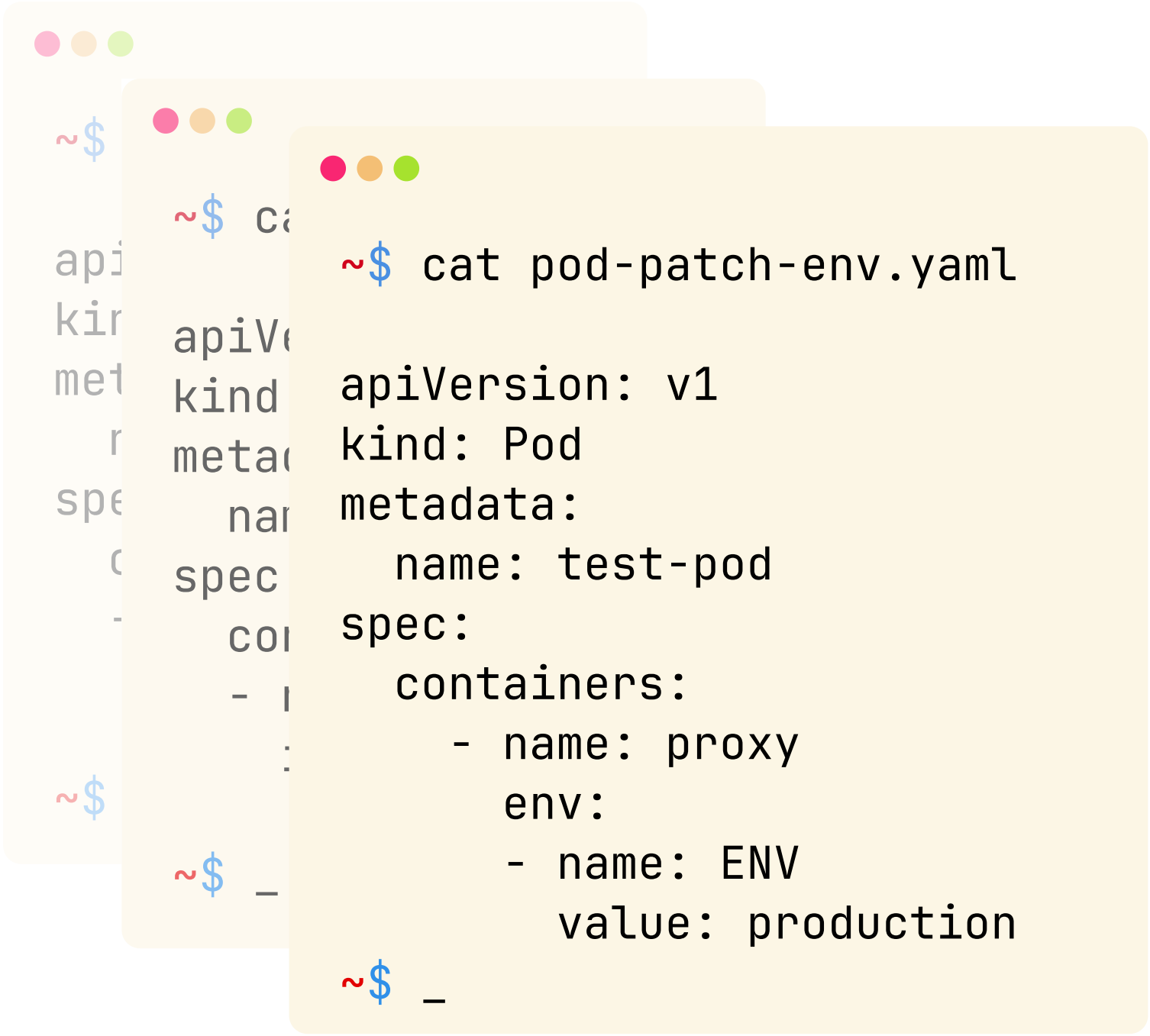
kustomize in practice

```
~$ cat app.yaml

apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
  - name: app
    image: busybox

~$ _
```


~$ cat proxy.yaml
apiVersion: v1
kind: Pod
metadata:
name: test-pod
spec:
containers:
- name: proxy
image: envoy
~$ _


~$ cat pod-patch-env.yaml
apiVersion: v1
kind: Pod
metadata:
name: test-pod
spec:
containers:
- name: proxy
env:
- name: ENV
value: production
~$ _

```
~$ kubectl kustomize prod

apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: proxy
      image: envoy
      env:
      - name: ENV
        value: production
    - name: app
      image: busybox
```

Recap

Recap

1. Search and replace strings
2. yq is YAML aware
3. yq merge
4. kustomize is Kubernetes YAML aware
5. kustomize patches and overlays

Recap

1. Search and replace strings
2. yq is YAML aware
3. yq merge
4. kustomize is Kubernetes YAML aware
5. kustomize patches and overlays

Recap

1. Search and replace strings
2. yq is YAML aware
3. yq merge
4. kustomize is Kubernetes YAML aware
5. kustomize patches and overlays

Recap

1. Search and replace strings
2. yq is YAML aware
3. yq merge
4. **kustomize is Kubernetes YAML aware**
5. kustomize patches and overlays

Recap

1. Search and replace strings
2. yq is YAML aware
3. yq merge
4. kustomize is Kubernetes YAML aware
5. **kustomize patches and overlays**

Helm

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: {{ .Values.env_name }}
~$ _
```

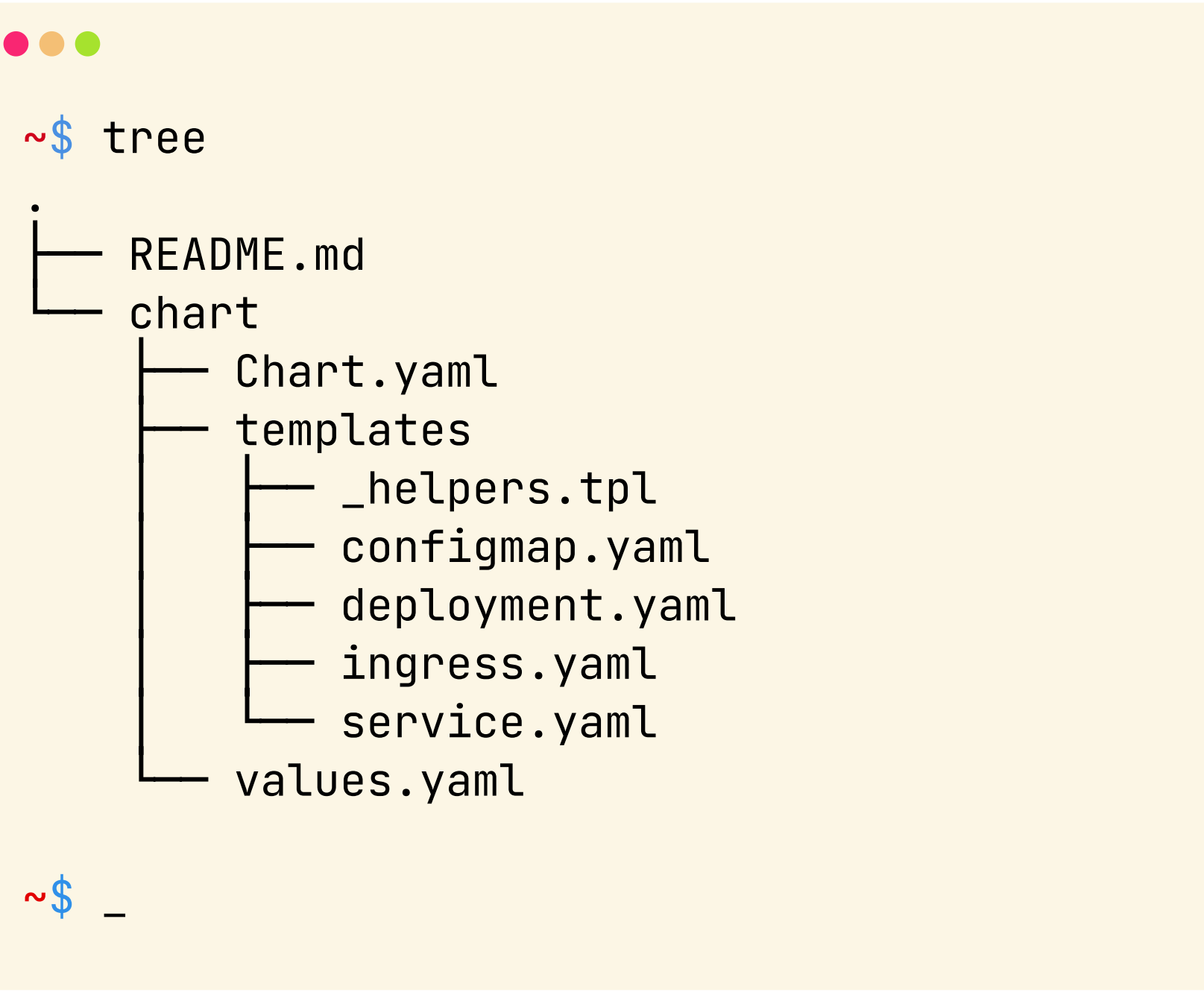
Go templating

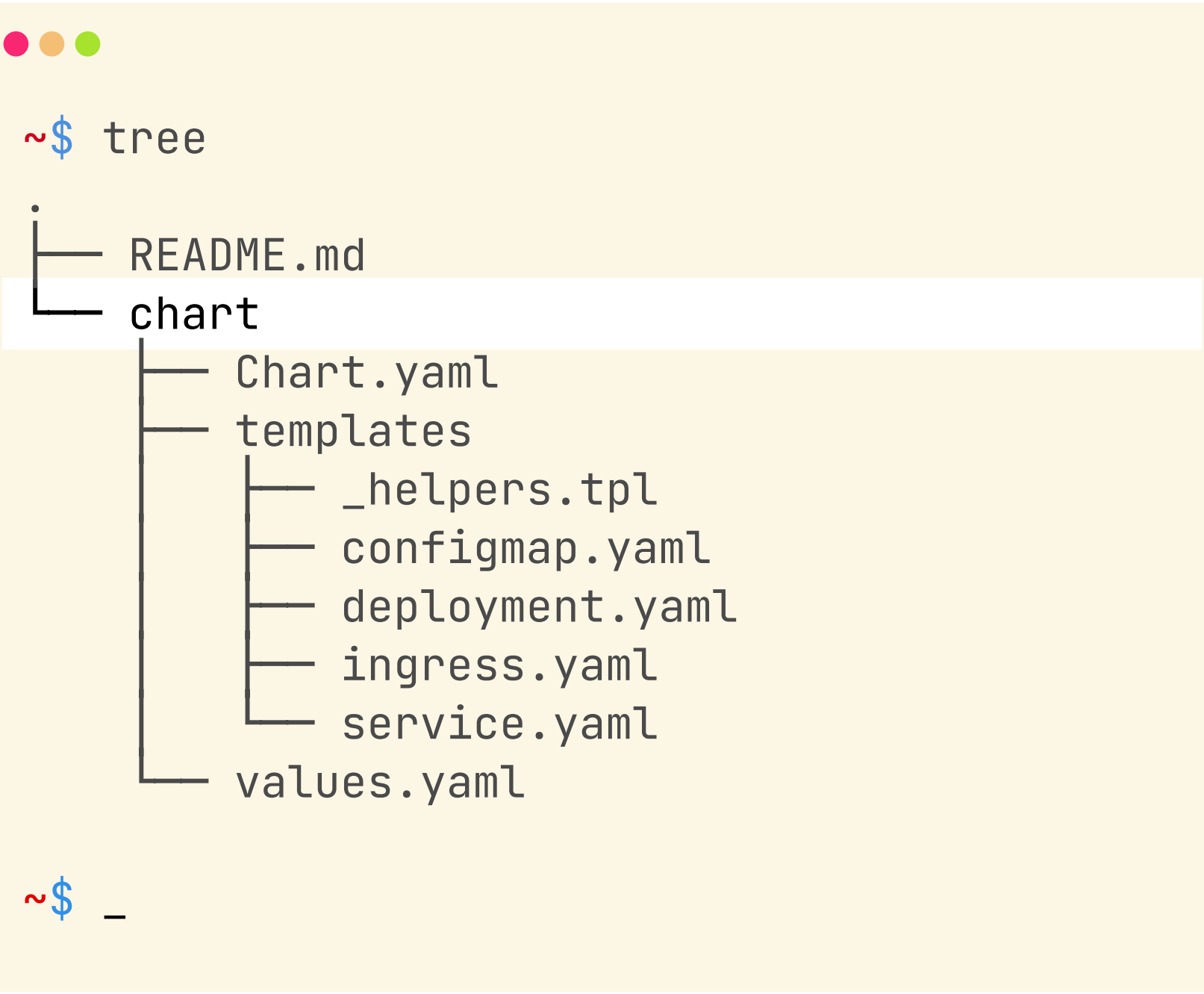
Sprig helpers

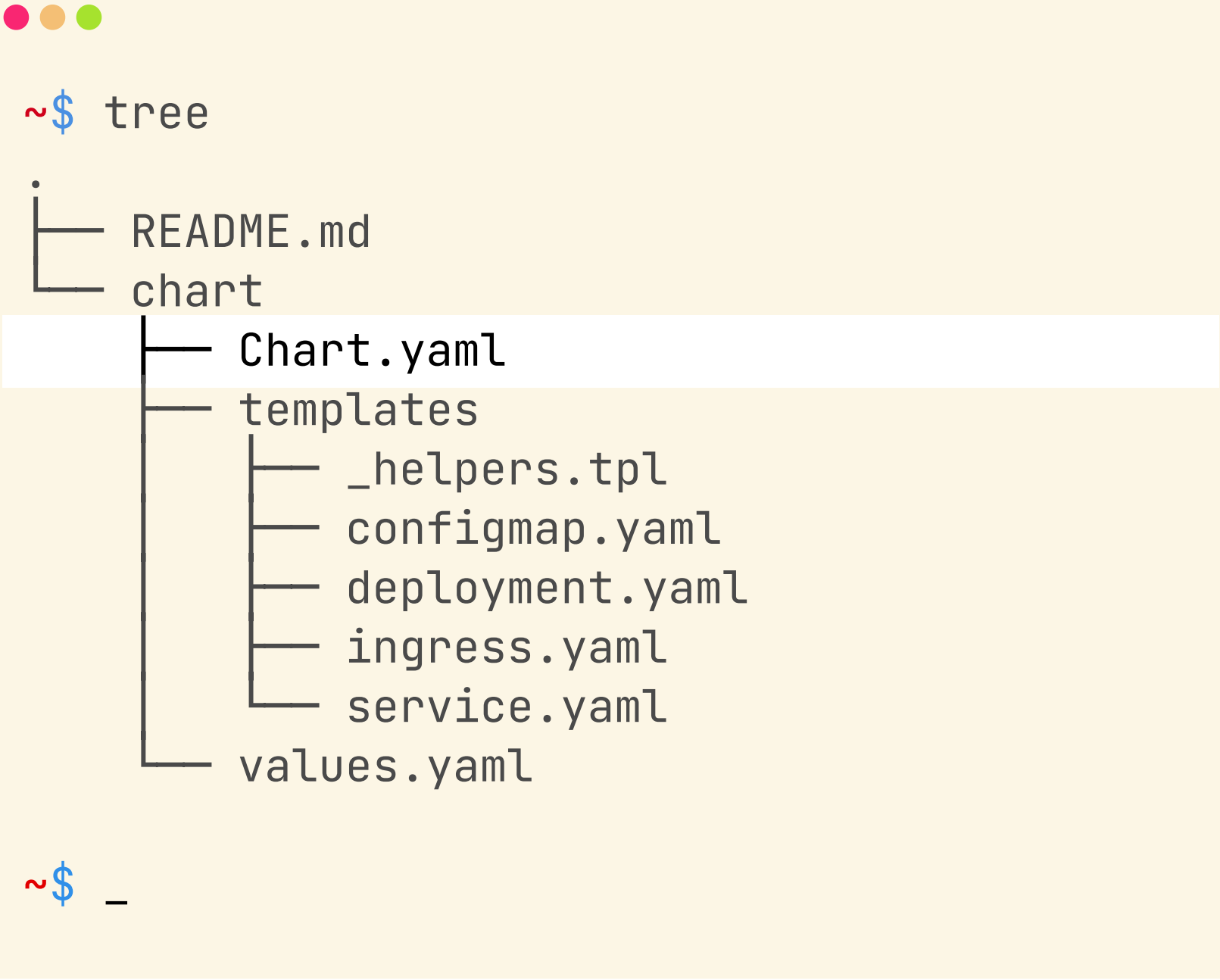
Go templating

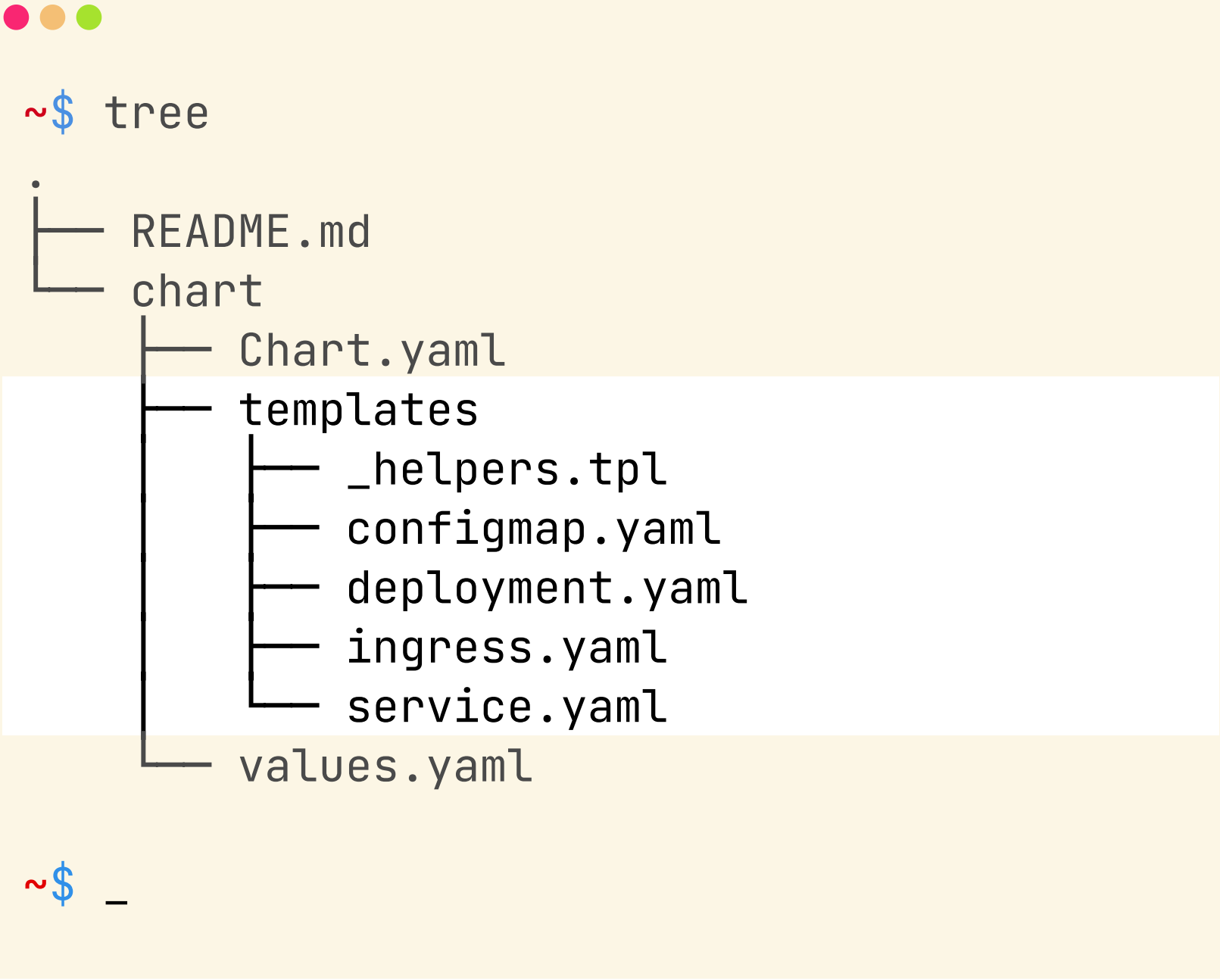
Sprig helpers

Helm charts


~$ tree
.
├── README.md
└── chart
    ├── Chart.yaml
    ├── templates
    │   ├── _helpers.tpl
    │   ├── configmap.yaml
    │   ├── deployment.yaml
    │   ├── ingress.yaml
    │   └── service.yaml
    └── values.yaml
~$ _


~$ tree
.
├── README.md
└── chart
    ├── Chart.yaml
    ├── templates
    │   ├── _helpers.tpl
    │   ├── configmap.yaml
    │   ├── deployment.yaml
    │   ├── ingress.yaml
    │   └── service.yaml
    └── values.yaml
~$ _


~$ tree
.
├── README.md
└── chart
    ├── Chart.yaml
    ├── templates
    │   ├── _helpers.tpl
    │   ├── configmap.yaml
    │   ├── deployment.yaml
    │   ├── ingress.yaml
    │   └── service.yaml
    └── values.yaml
~$ _


~$ tree
.
├── README.md
└── chart
    ├── Chart.yaml
    ├── templates
    │   ├── _helpers.tpl
    │   ├── configmap.yaml
    │   ├── deployment.yaml
    │   ├── ingress.yaml
    │   └── service.yaml
    └── values.yaml
~$ _

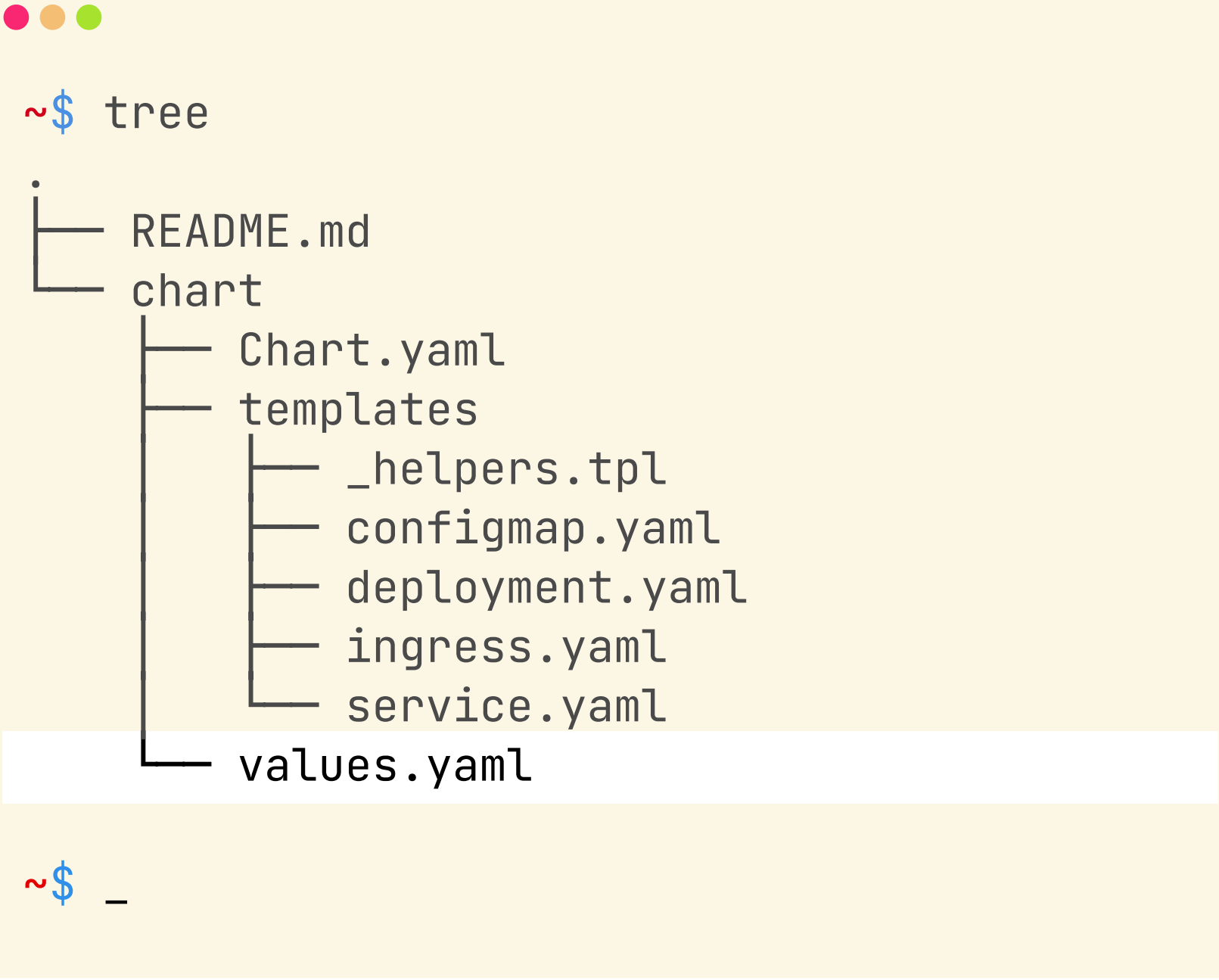

~$ tree
.
├── README.md
└── chart
    ├── Chart.yaml
    ├── templates
    │   ├── _helpers.tpl
    │   ├── configmap.yaml
    │   ├── deployment.yaml
    │   ├── ingress.yaml
    │   └── service.yaml
    └── values.yaml
~$ _

Chart.yaml

```
~$ cat chart/Chart.yaml

apiVersion: v2
name: node-hello-world
version: 1.0.0

~$ _
```

values.yaml

```
~$ cat chart/values.yaml

env_name: 'production'
image: learnk8s/node-hello-world
replicas: 3

~$ _
```

```
~$ cat chart/values.yaml
```

```
env_name: 'production'
image: learnk8s/node-hello-world
replicas: 3
```

```
~$ _
```

Templates for resources

```
~$ cat chart/templates/pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: {{ .Values.env_name }}

~$ _
```

```
~$ cat chart/templates/pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: test-pod
spec:
  containers:
    - name: test-container
      image: k8s.gcr.io/busybox
      env:
        - name: ENV
          value: {{ .Values.env_name }}

~$ _
```

```
~$ cat chart/values.yaml
```

```
env_name: 'production'
image: learnk8s/node-hello-world
replicas: 3
```

```
~$ _
```



```
~$ helm install <release-name> <folder> \
  --set env_name=dev

~$ _
```

Managing releases

$ helm install <name> <folder>

Creates a release

$ helm ls

Lists releases



```
~$ helm ls

NAME          NAMESPACE REVISION STATUS    CHAR
dev-mariadb default   1         deployed maria
dev-api     default   1         deployed maria
dev-fe       default    1         deployed mari
dev-worker1  default    1         deployed mari

~$ _
```

$ helm delete <release name>

Deletes a release

Rollbacks

Revision	What happened
1	Installed the chart

Revision	What happened
1	Installed the chart
2	Upgraded to v1.0.0

Revision	What happened
1	Installed the chart
2	Upgraded to v1.0.0
3	Upgraded to v2.0.0

Revision	What happened
1	Installed the chart
2	Upgraded to v1.0.0
3	Upgraded to v2.0.0

```
$ helm rollback \
  <release> <revision>
```

Revision	What happened
1	Installed the chart
2	Upgraded to v1.0.0
3	Upgraded to v2.0.0
4	Rolled back to 2

$ helm history <release name>

Shows release history

Sharing and discovering charts



```
~$ helm repo add prom \
   https://prometheus-community.github.io/helm-charts

~$ helm search repo prom/prometheus --version 29.36.1

NAME              CHART VERSION  APP VERSION
prom/prometheus   29.36.1        v3.15.0

~$ _
```

```
~$ helm install prometheus-dev \
   prom/prometheus --version 29.36.1

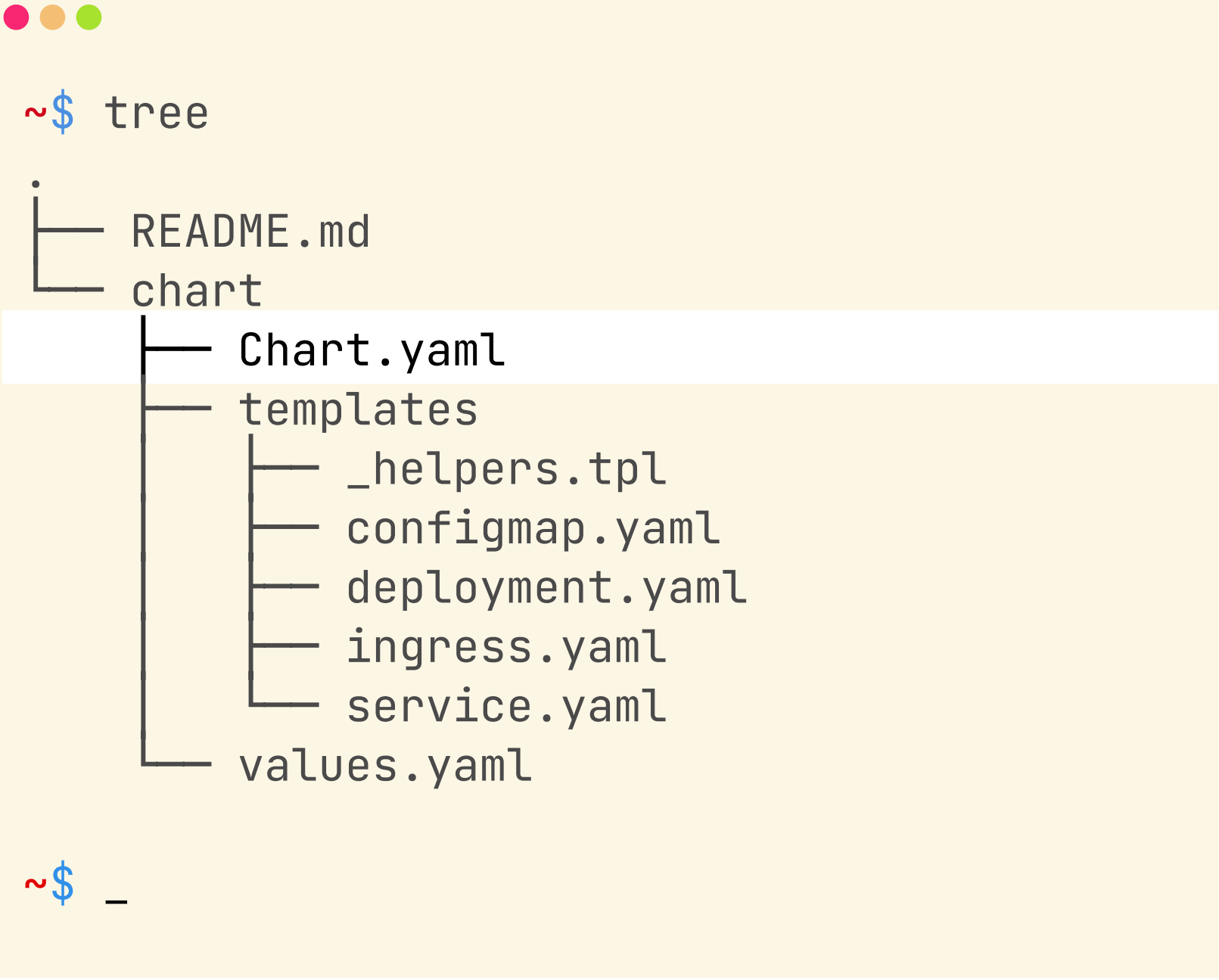
NAME: prometheus-dev
NAMESPACE: default
STATUS: deployed
REVISION: 1

~$ _
```



Helm chart repository

Dependencies on other charts


~$ tree
.
├── README.md
└── chart
    ├── Chart.yaml
    ├── templates
    │   ├── _helpers.tpl
    │   ├── configmap.yaml
    │   ├── deployment.yaml
    │   ├── ingress.yaml
    │   └── service.yaml
    └── values.yaml
~$ _

```
~$ cat chart/Chart.yaml
apiVersion: v2
name: node-hello-world
version: 1.0.0
dependencies:
  - name: apache
    version: 1.2.3
    repository: http://example.com/charts
  - name: mysql
    version: 3.2.1
    repository: http://example.com/charts

~$ _
```

```
~$ cat chart/Chart.yaml
apiVersion: v2
name: node-hello-world
version: 1.0.0
dependencies:
  - name: apache
    version: 1.2.3
    repository: http://example.com/charts
  - name: mysql
    version: 3.2.1
    repository: http://example.com/charts

~$ _
```

```
~$ cat chart/Chart.yaml
apiVersion: v2
name: node-hello-world
version: 1.0.0
dependencies:
  - name: apache
    version: 1.2.3
    repository: http://example.com/charts
  - name: mysql
    version: 3.2.1
    repository: http://example.com/charts

~$ helm dependency update chart
```

Field ownership



```
last-applied-configuration
replicas: 3
---
apiVersion: apps/v1
kind: Deployment
spec:
  replicas: 3
```

matching





HPA

```
last-applied-configuration
replicas: 3
---
apiVersion: apps/v1
kind: Deployment
spec:
  replicas: 5
```

/scale to 5





kubectl apply -f deployment.yaml

```
last-applied-configuration
replicas: 3
---
apiVersion: apps/v1
kind: Deployment
spec:
  replicas: 5
```

stale





kubectl apply -f deployment.yaml

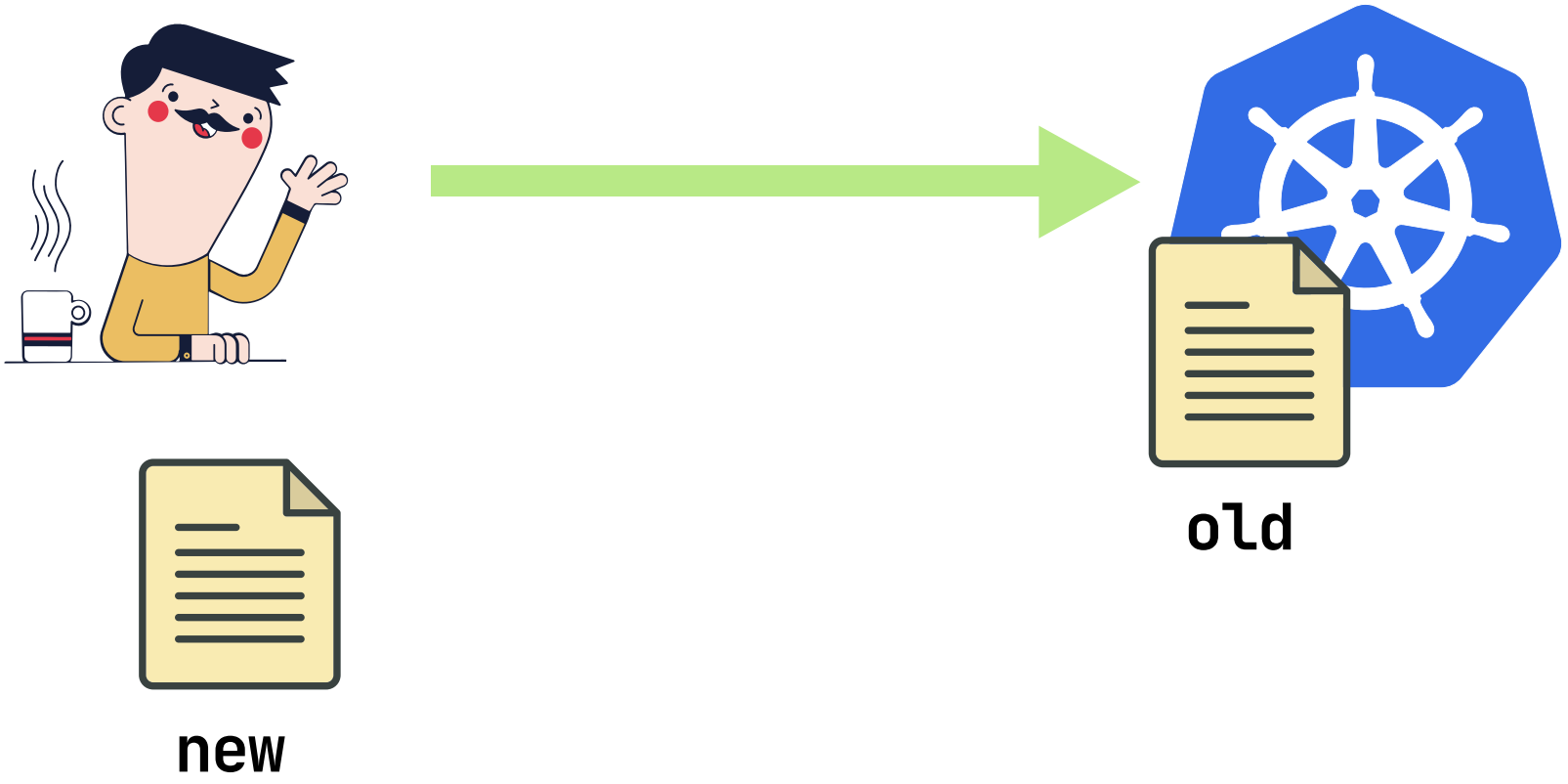
```
last-applied-configuration
replicas: 3
---
apiVersion: apps/v1
kind: Deployment
spec:
  replicas: 3
```

reverted

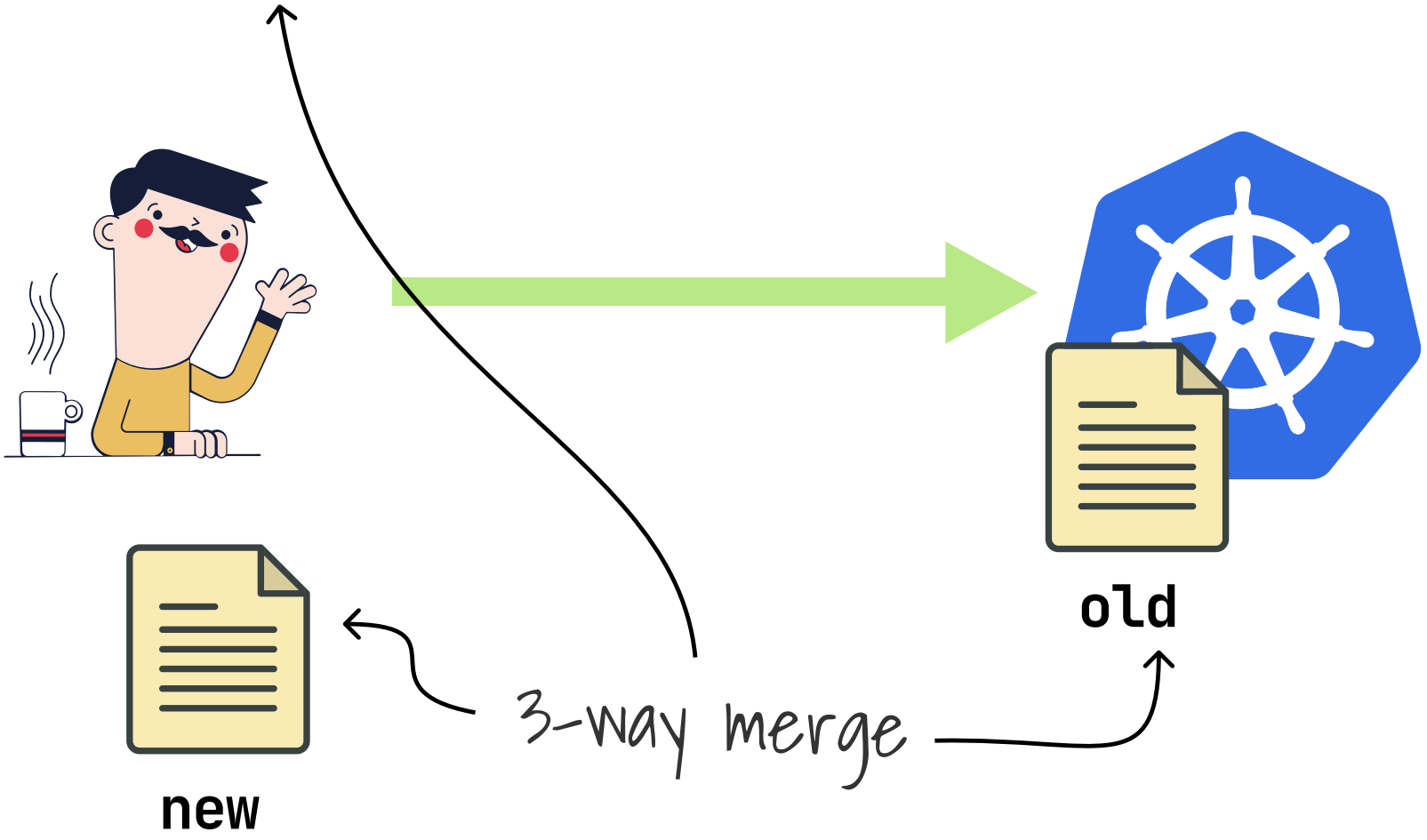


Client vs server

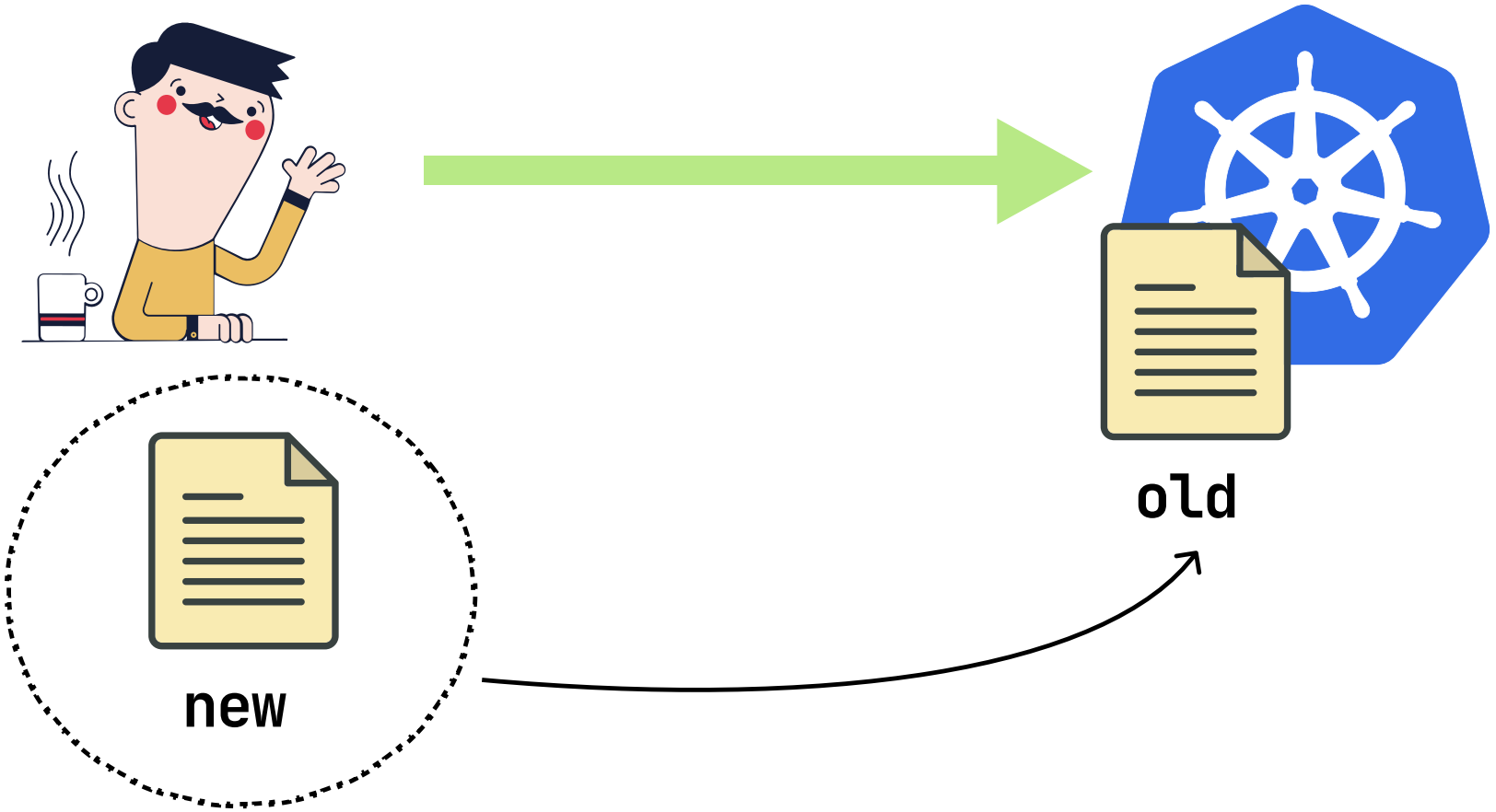
kubectl apply -f deployment.yaml



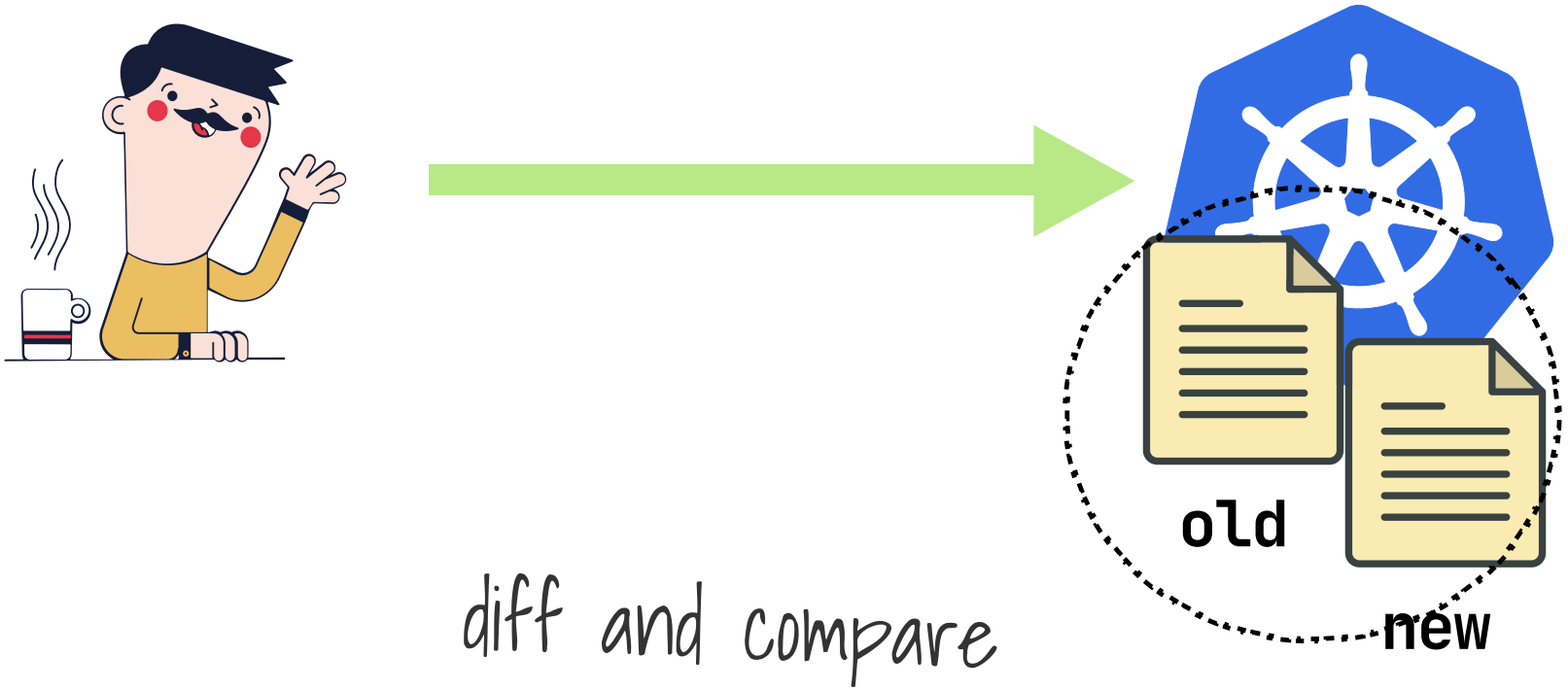
kubectl apply -f deployment.yaml



kubectl apply --server-side -f deployment.yaml

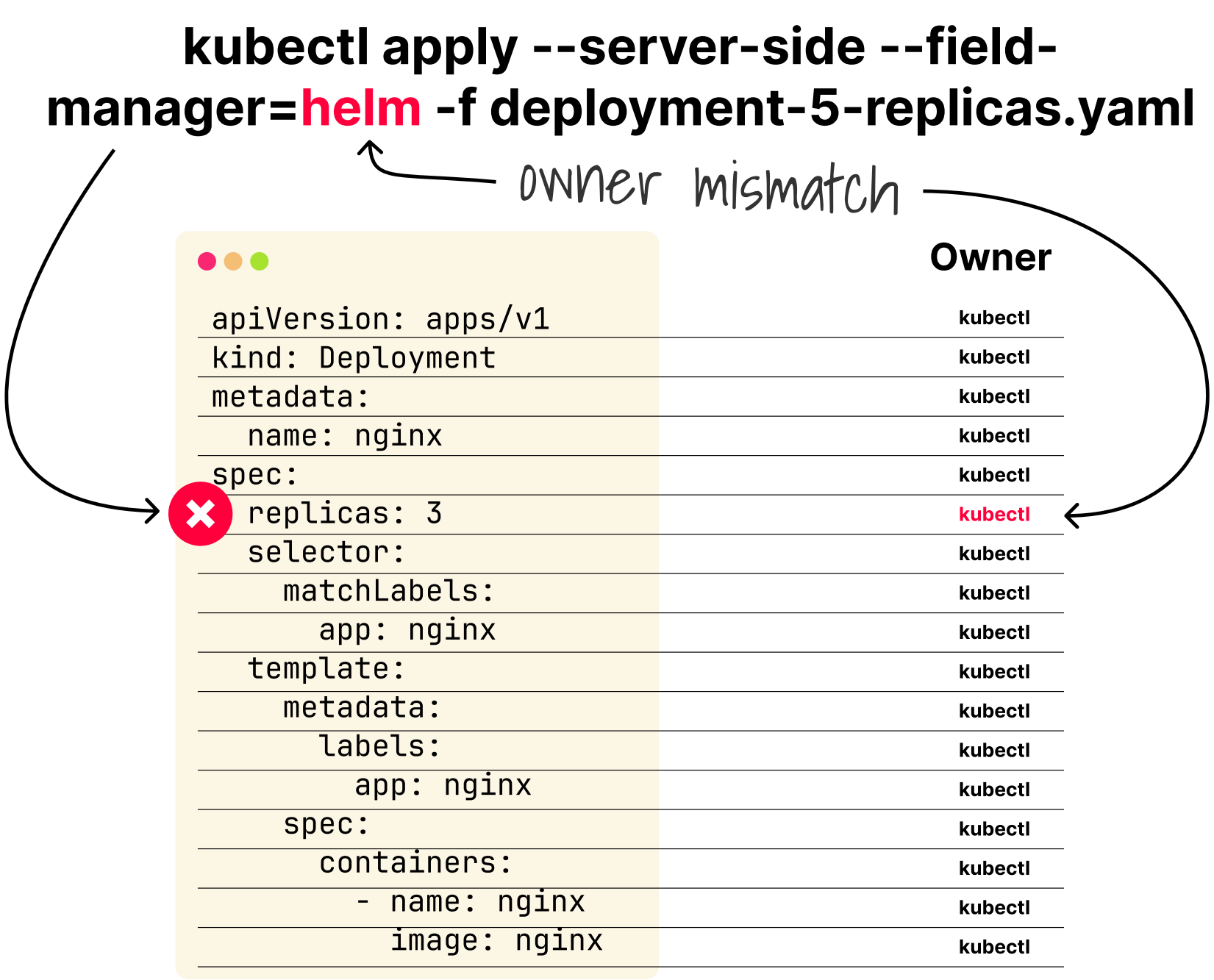


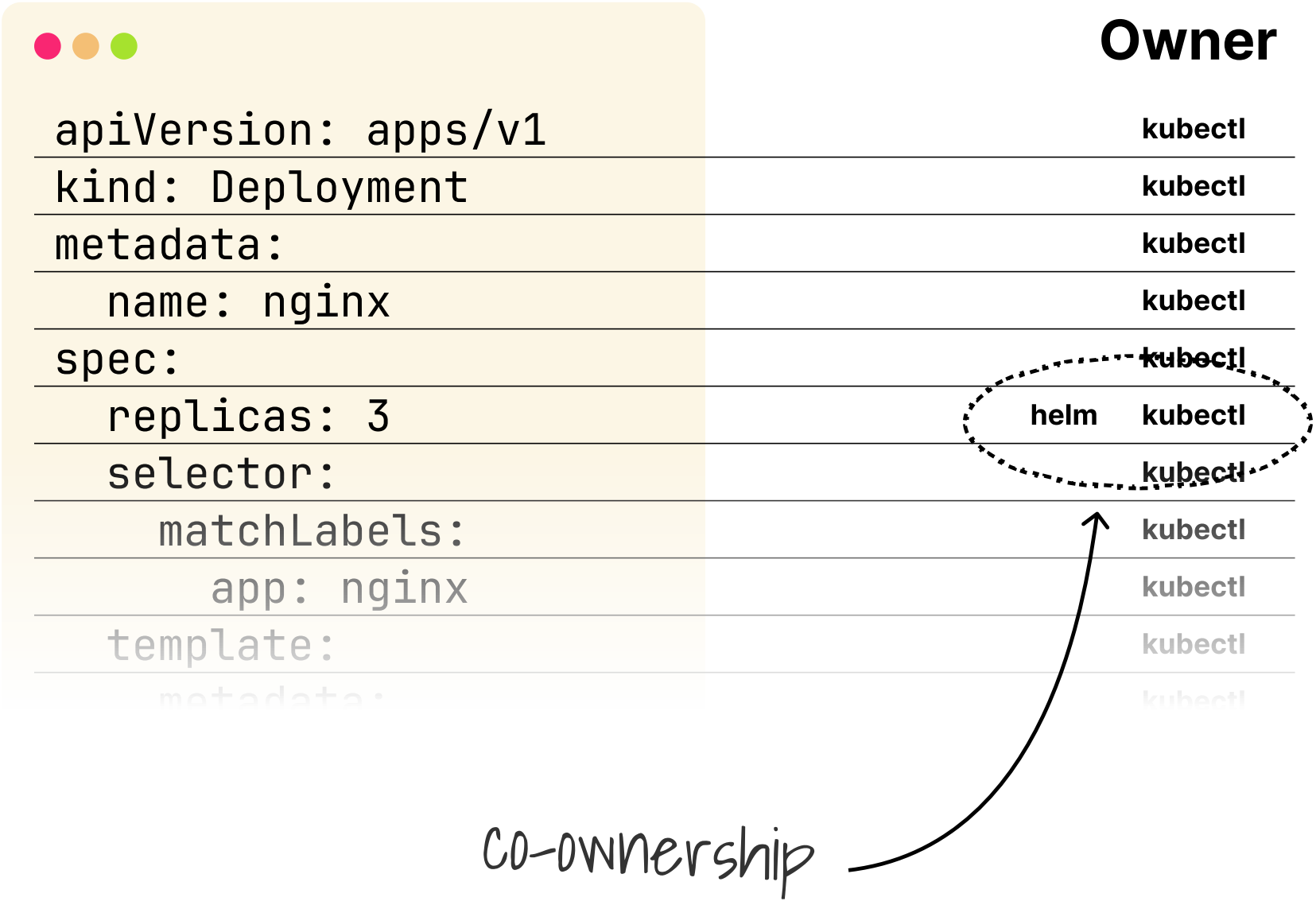
kubectl apply --server-side -f deployment.yaml

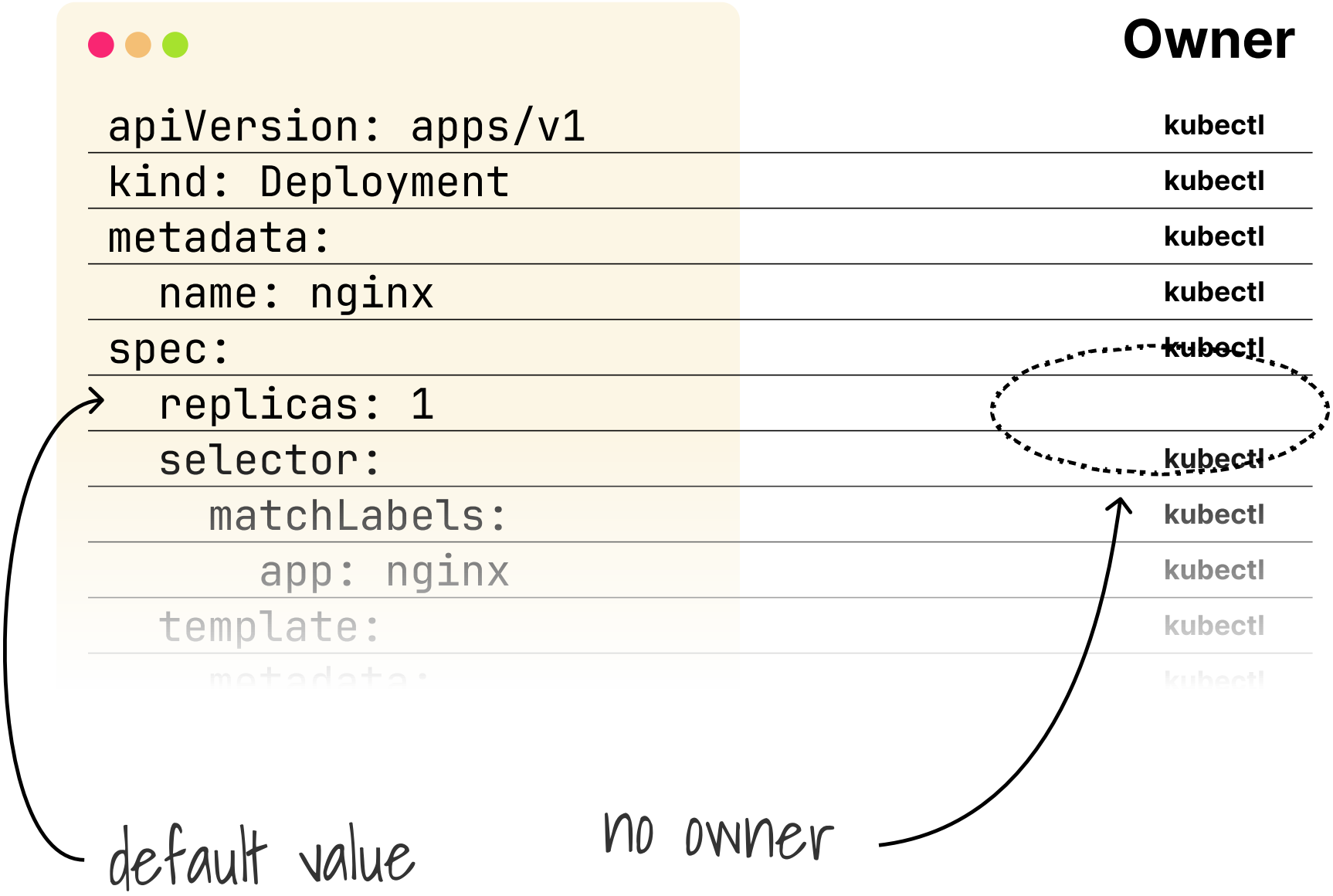


Server-Side Apply

	Owner
`apiVersion: apps/v1`	kubectl
`kind: Deployment`	kubectl
`metadata:`	kubectl
`  name: nginx`	kubectl
`spec:`	kubectl
`  replicas: 3`	kubectl
`  selector:`	kubectl
`    matchLabels:`	kubectl
`      app: nginx`	kubectl
`  template:`	kubectl
`    metadata:`	kubectl
`      labels:`	kubectl
`        app: nginx`	kubectl
`    spec:`	kubectl
`      containers:`	kubectl
`        - name: nginx`	kubectl
`          image: nginx`	kubectl


kubectl apply --server-side --field-manager=helm -f deployment-5-replicas.yaml
owner mismatch
Owner
apiVersion: apps/v1 kubectl
kind: Deployment kubectl
metadata: kubectl
name: nginx kubectl
spec: kubectl
replicas: 3 kubectl
selector: kubectl
matchLabels: kubectl
app: nginx kubectl
template: kubectl
metadata: kubectl
labels: kubectl
app: nginx kubectl
spec: kubectl
containers: kubectl
- name: nginx kubectl
image: nginx kubectl


Owner
apiVersion: apps/v1
kind: Deployment
metadata:
name: nginx
spec:
replicas: 3
selector:
matchLabels:
app: nginx
template:
kubectl
kubectl
kubectl
kubectl
kubectl
helm
kubectl
kubectl
kubectl
kubectl
kubectl
Co-ownership


Owner
apiVersion: apps/v1
kind: Deployment
metadata:
name: nginx
spec:
replicas: 1
selector:
matchLabels:
app: nginx
template:
kubectl
kubectl
kubectl
kubectl
kubectl
kubectl
kubectl
kubectl
default value
no owner

Helm 4 is SSA-enabled by default

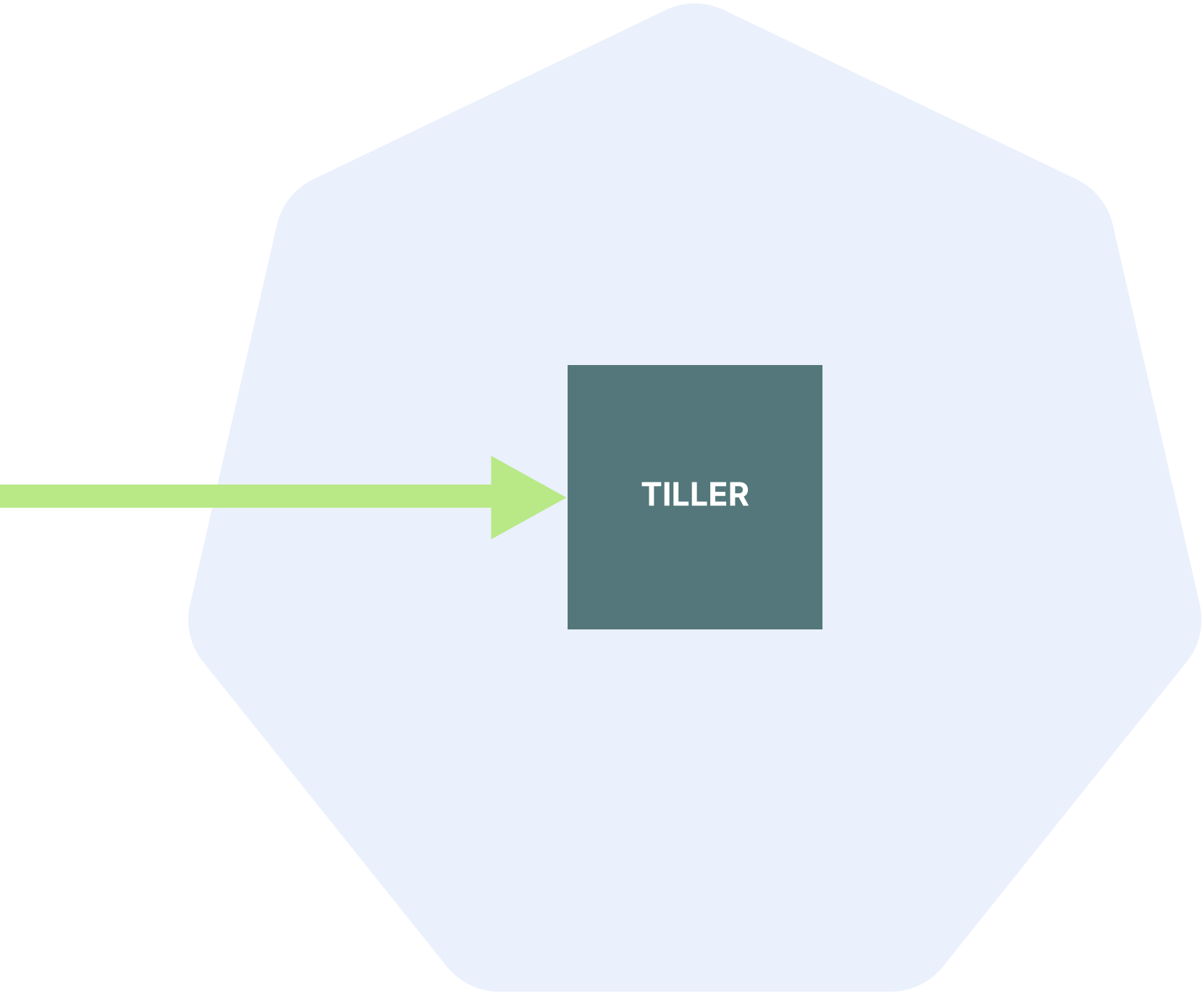
Server-Side Apply

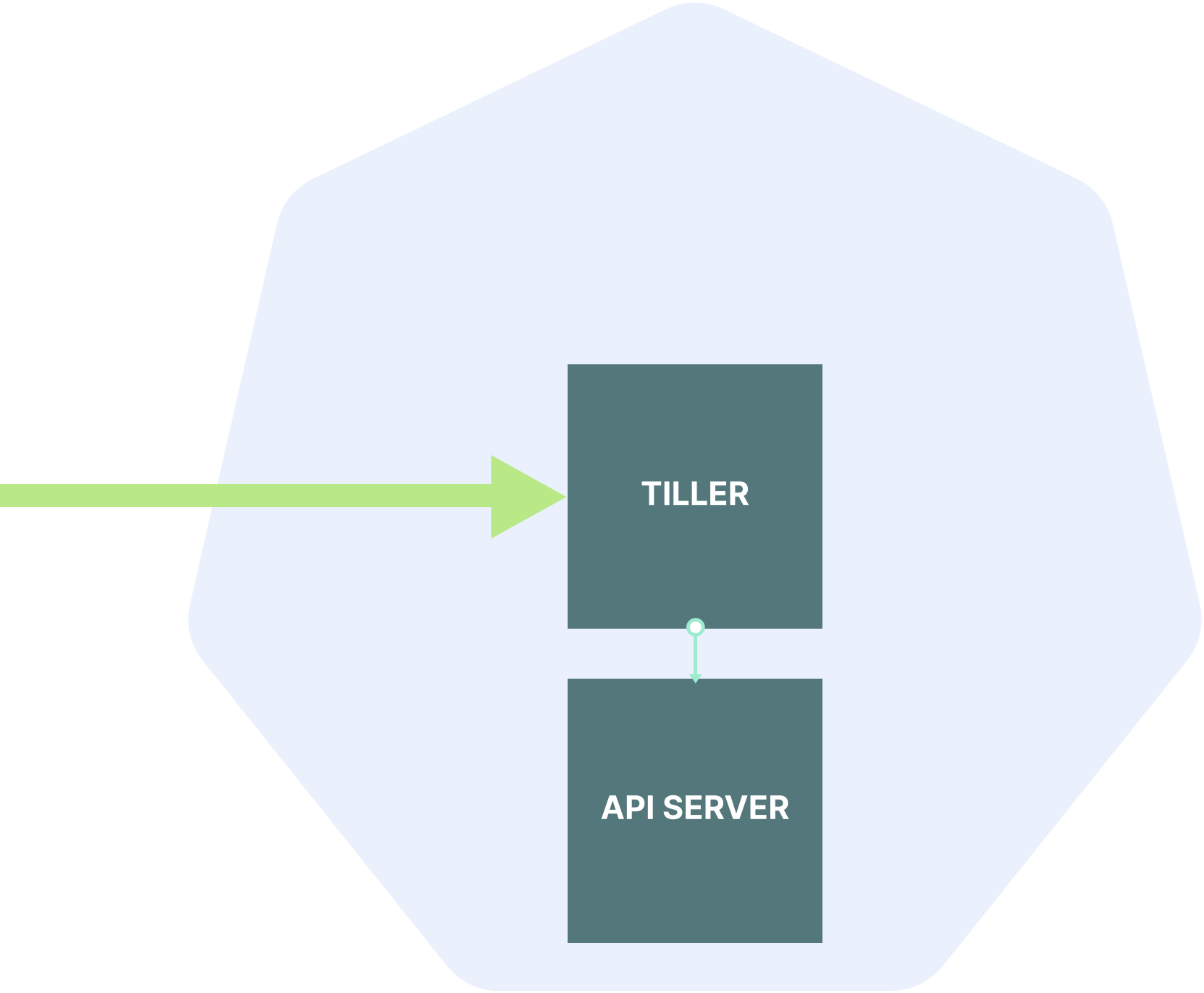
Helm 2 architecture

Deploy a templated pod






TILLER


TILLER
API SERVER

Helm 2 architecture

Reach information

Manage releases

Rollbacks

Helm 2 architecture

Reach information

Manage releases

Rollbacks

Helm 2 architecture

Reach information ✓

Manage releases ✓

Rollbacks ✓

Helm 2 architecture

Permissions ✖

Controller

Helm 2 architecture

Permissions ✗

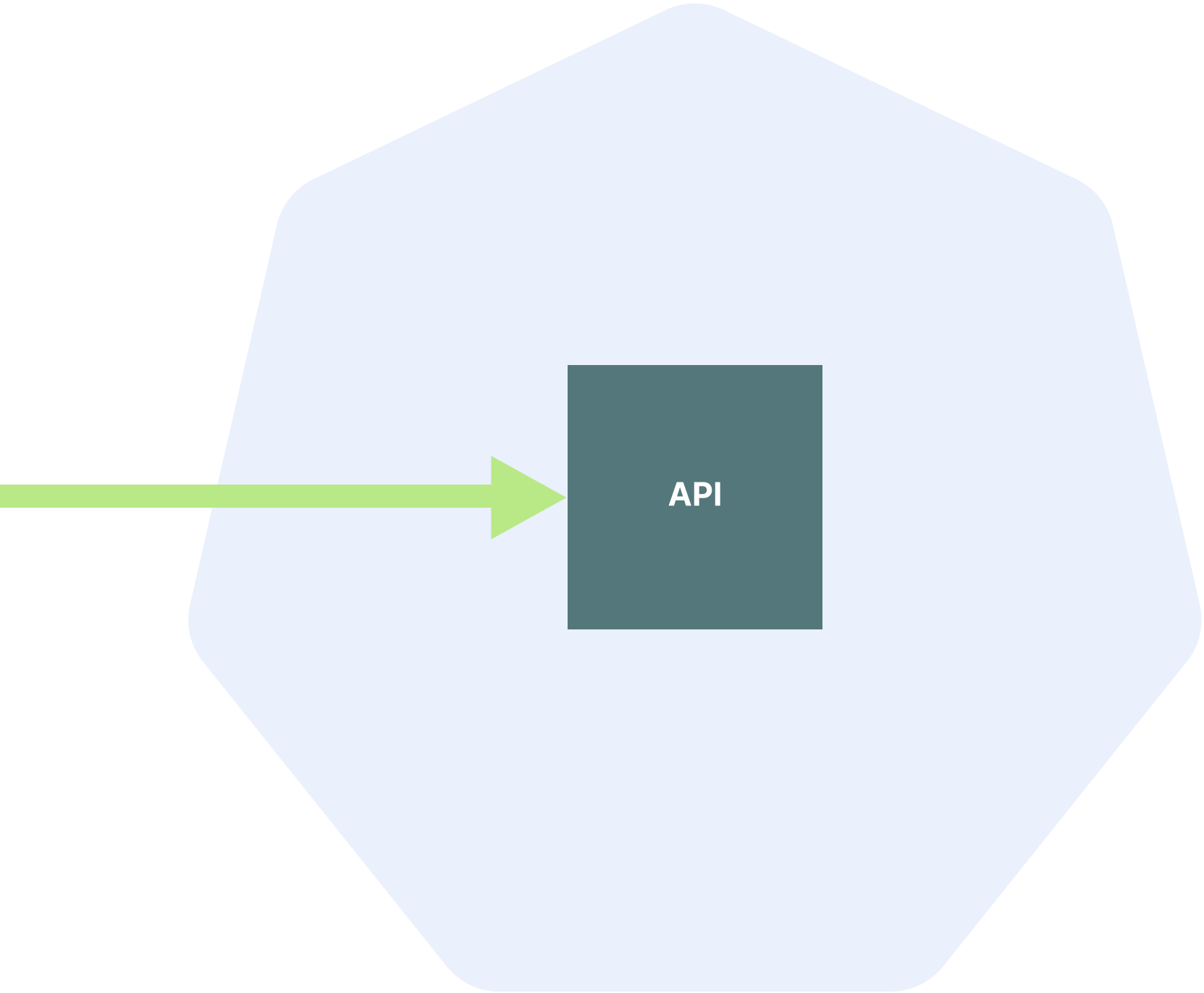
Controller ✗

Helm 3+ architecture

Deploy a templated pod






API

Helm 3+

No tiller 

Helm 3+

No tiller

State in secrets

Helm 3+

No tiller

State in secrets

Permissions: kubeconfig

Helm shortcomings

```
~$ cat configmap.yaml

apiVersion: v1
kind: ConfigMap
metadata:
  name: {{ template "name" . }}
  labels:
    app: {{ template "name" . }}
    chart: {{ .Chart.Name }}-{{ .Chart.Version | replace "+" "_" }}
    release: {{ quote .Release.Name }}
    heritage: {{ quote .Release.Service }}
    name: {{ template "name" . }}
data:
  logback.xml: |-
    {{- .Files.Get "logback.xml" | nindent 4 }}
  application.yml: |-
    {{- include "application.yml" . | nindent 4 }}

~$ _
```

```
~$ cat configmap.yaml

apiVersion: v1
kind: ConfigMap
metadata:
  name: {{ template "name" . }}
  labels:
    app: {{ template "name" . }}
    chart: {{ .Chart.Name }}-{{ .Chart.Version | replace "+" "_" }}
    release: {{ quote .Release.Name }}
    heritage: {{ quote .Release.Service }}
    name: {{ template "name" . }}
data:
  logback.xml: |-
    {{- .Files.Get "logback.xml" | nindent 4 }}
  application.yml: |-
    {{- include "application.yml" . | nindent 4 }}

~$ _
```

```
~$ cat chart/values.yaml
```

```
env_name: 'production'
image: learnk8s/node-hello-world
replicas: 3
```

```
~$ _
```

Recap

Recap

1. Helm templating
2. Helm charts
3. Helm release management
4. Helm architecture
5. Helm shortcomings

Recap

1. Helm templating
2. **Helm charts**
3. Helm release management
4. Helm architecture
5. Helm shortcomings

Recap

1. Helm templating
2. Helm charts
3. **Helm release management**
4. Helm architecture
5. Helm shortcomings

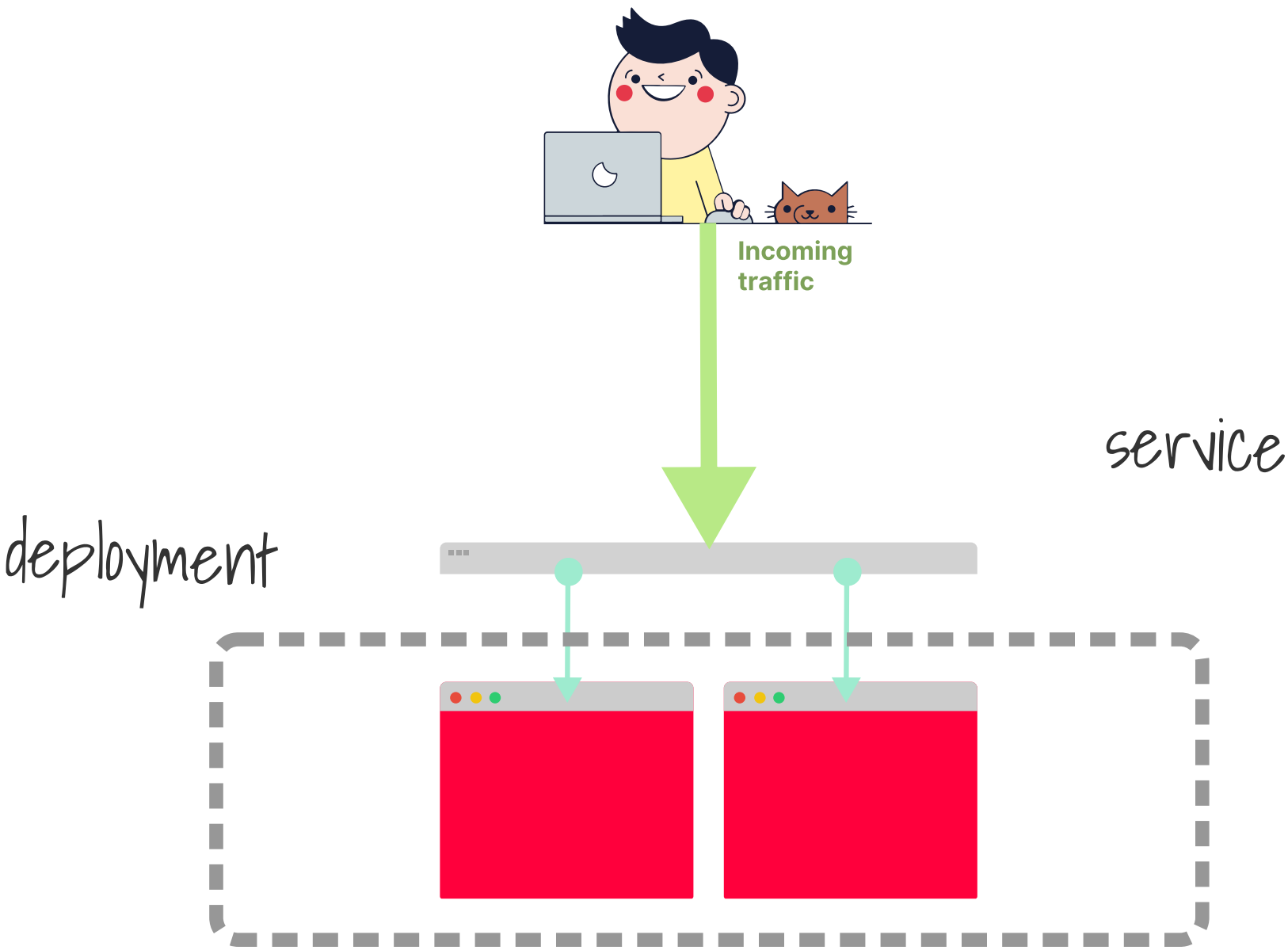
Recap

1. Helm templating
2. Helm charts
3. Helm release management
4. **Helm architecture**
5. Helm shortcomings

Recap

1. Helm templating
2. Helm charts
3. Helm release management
4. Helm architecture
5. **Helm shortcomings**

Better together


Incoming
traffic
service
deployment

service

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx
spec:
  selector:
    app: nginx
  ports:
    - port: 80
      targetPort: 80
```

deployment

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:1.27
```

Better together

Kustomization

Better together

Kustomization

Helm chart

Better together

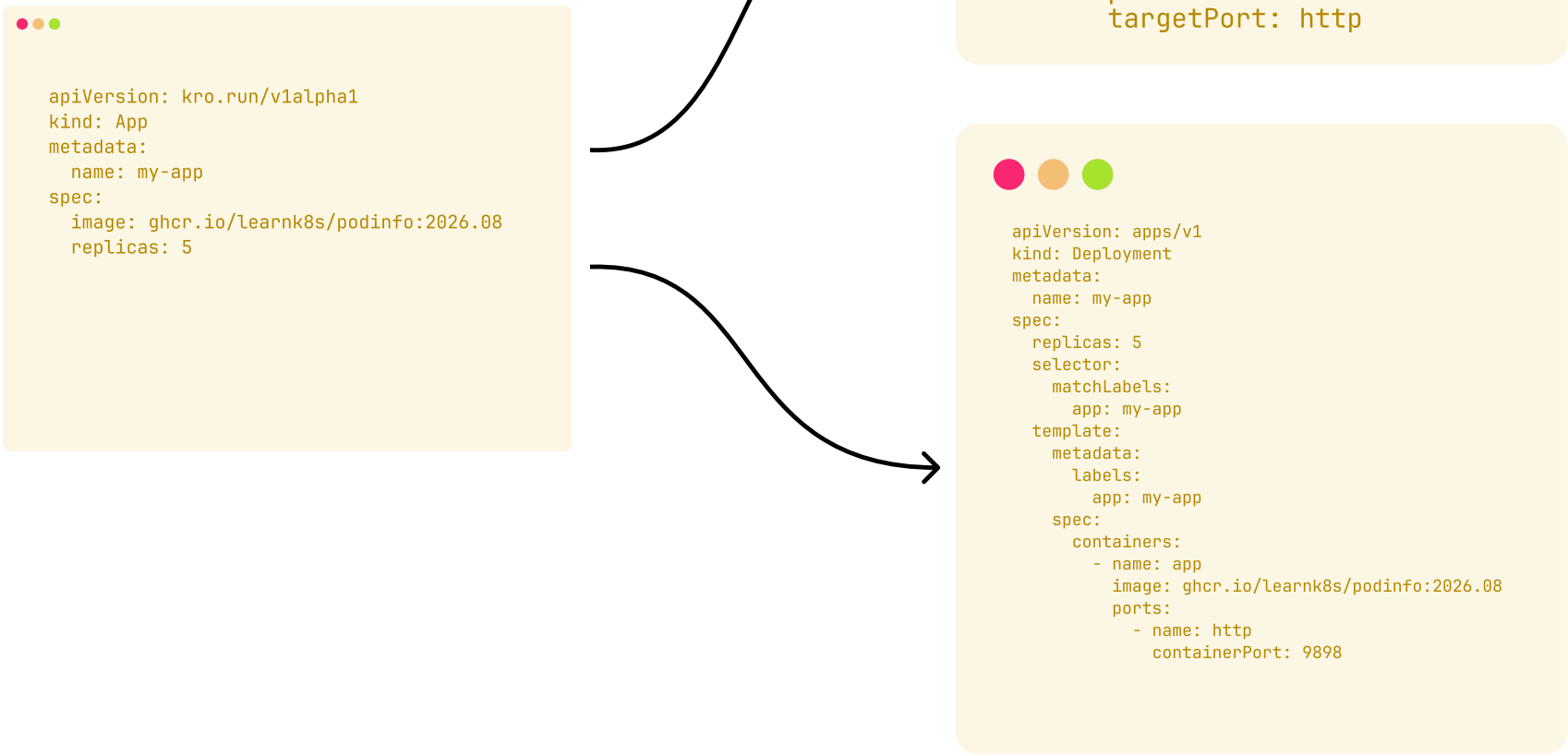
Kustomization

Helm chart

CRD

KRO

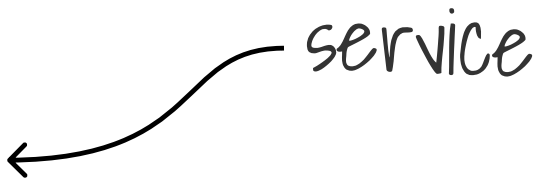
```yaml
apiVersion: kro.run/v1alpha1
kind: App
metadata:
  name: my-app
spec:
  image: ghcr.io/learnk8s/podinfo:2026.08
  replicas: 5
```


apiVersion: kro.run/v1alpha1
kind: App
metadata:
name: my-app
spec:
image: ghcr.io/learnk8s/podinfo:2026.08
replicas: 5
apiVersion: v1
kind: Service
metadata:
name: my-app
spec:
selector:
app: my-app
ports:
- name: http
port: 80
targetPort: http
apiVersion: apps/v1
kind: Deployment
metadata:
name: my-app
spec:
replicas: 5
selector:
matchLabels:
app: my-app
template:
metadata:
labels:
app: my-app
spec:
containers:
- name: app
image: ghcr.io/learnk8s/podinfo:2026.08
ports:
- name: http
containerPort: 9898

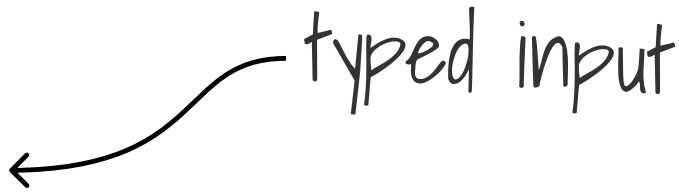
```yaml
apiVersion: kro.run/v1alpha1
kind: ResourceGraphDefinition
metadata:
  name: apps.course
spec:
  schema:
    apiVersion: v1alpha1
    kind: App
    spec:
      image: string | default="ghcr.io/learnk8s/podinfo:2026.08"
      replicas: integer | default=1 minimum=1 maximum=5
  resources:
    - id: deployment
      template:
        apiVersion: apps/v1
        kind: Deployment
        metadata:
          name: ${schema.metadata.name}
        spec:
          replicas: ${schema.spec.replicas}
          selector:
            matchLabels:
              app: ${schema.metadata.name}
          template:
            metadata:
              labels:
                app: ${schema.metadata.name}
            spec:
              containers:
                - name: app
                  image: ${schema.spec.image}
                  ports:
                    - name: http
                      containerPort: 9898
    - id: service
      template:
        apiVersion: v1
        kind: Service
        metadata:
          name: ${deployment.metadata.name}
        spec:
          selector: ${deployment.spec.selector.matchLabels}
          ports:
            - name: http
              port: 80
              targetPort: http
```



```yaml
apiVersion: kro.run/v1alpha1
kind: ResourceGraphDefinition
metadata:
  name: apps.course
spec:
  schema:
    apiVersion: v1alpha1
    kind: App
    spec:
      image: string | default="ghcr.io/learnk8s/podinfo:2026.08"
      replicas: integer | default=1 minimum=1 maximum=5
  resources:
    - id: deployment
      template:
        apiVersion: apps/v1
        kind: Deployment
        metadata:
          name: ${schema.metadata.name}
        spec:
          replicas: ${schema.spec.replicas}
          selector:
            matchLabels:
              app: ${schema.metadata.name}
          template:
            metadata:
              labels:
                app: ${schema.metadata.name}
            spec:
              containers:
                - name: app
                  image: ${schema.spec.image}
                  ports:
                    - name: http
                      containerPort: 9898
    - id: service
      template:
        apiVersion: v1
        kind: Service
        metadata:
          name: ${deployment.metadata.name}
        spec:
          selector: ${deployment.spec.selector.matchLabels}
          ports:
            - name: http
              port: 80
              targetPort: http
```



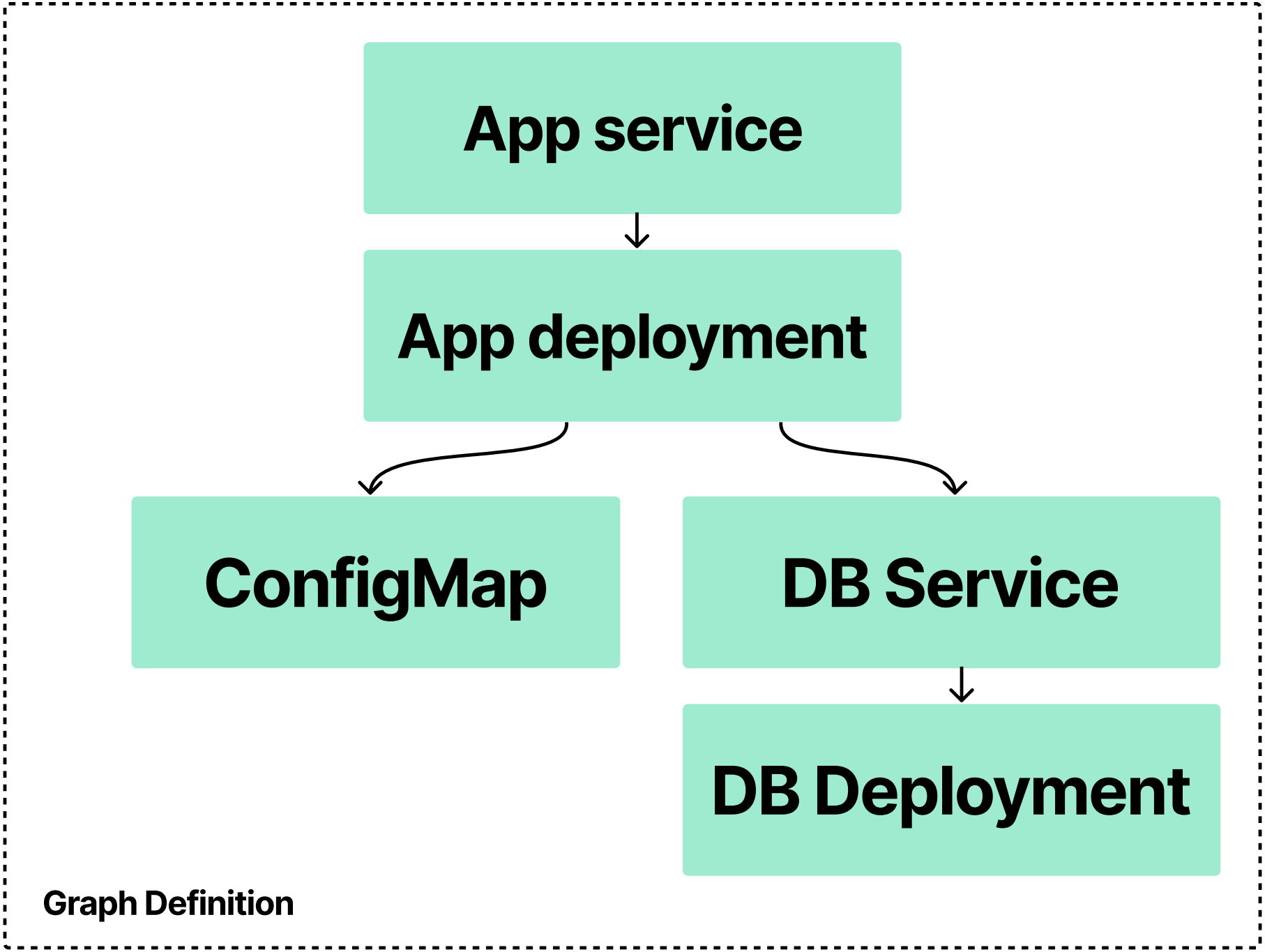
```yaml
apiVersion: kro.run/v1alpha1
kind: ResourceGraphDefinition
metadata:
  name: apps.course
spec:
  schema:
    apiVersion: v1alpha1
    kind: App
    spec:
      image: string | default="ghcr.io/learnk8s/podinfo:2026.08"
      replicas: integer | default=1 minimum=1 maximum=5
  resources:
    - id: deployment
      template:
        apiVersion: apps/v1
        kind: Deployment
        metadata:
          name: ${schema.metadata.name}
        spec:
          replicas: ${schema.spec.replicas}
          selector:
            matchLabels:
              app: ${schema.metadata.name}
          template:
            metadata:
              labels:
                app: ${schema.metadata.name}
            spec:
              containers:
                - name: app
                  image: ${schema.spec.image}
                  ports:
                    - name: http
                      containerPort: 9898
    - id: service
      template:
        apiVersion: v1
        kind: Service
        metadata:
          name: ${deployment.metadata.name}
        spec:
          selector: ${deployment.spec.selector.matchLabels}
          ports:
            - name: http
              port: 80
              targetPort: http
```



```yaml
apiVersion: kro.run/v1alpha1
kind: ResourceGraphDefinition
metadata:
  name: apps.course
spec:
  schema:
    apiVersion: v1alpha1
    kind: App
    spec:
      image: string | default="ghcr.io/learnk8s/podinfo:2026.08"
      replicas: integer | default=1 minimum=1 maximum=5
  resources:
    - id: deployment
      template:
        apiVersion: apps/v1
        kind: Deployment
        metadata:
          name: ${schema.metadata.name}
        spec:
          replicas: ${schema.spec.replicas}
          selector:
            matchLabels:
              app: ${schema.metadata.name}
          template:
            metadata:
              labels:
                app: ${schema.metadata.name}
            spec:
              containers:
                - name: app
                  image: ${schema.spec.image}
                  ports:
                    - name: http
                      containerPort: 9898
    - id: service
      template:
        apiVersion: v1
        kind: Service
        metadata:
          name: ${deployment.metadata.name}
        spec:
          selector: ${deployment.spec.selector.matchLabels}
          ports:
            - name: http
              port: 80
              targetPort: http
```



Graph dependencies


App service
App deployment
ConfigMap
DB Service
DB Deployment
Graph Definition

App service

App deployment

ConfigMap

DB Service

DB Deployment

App service

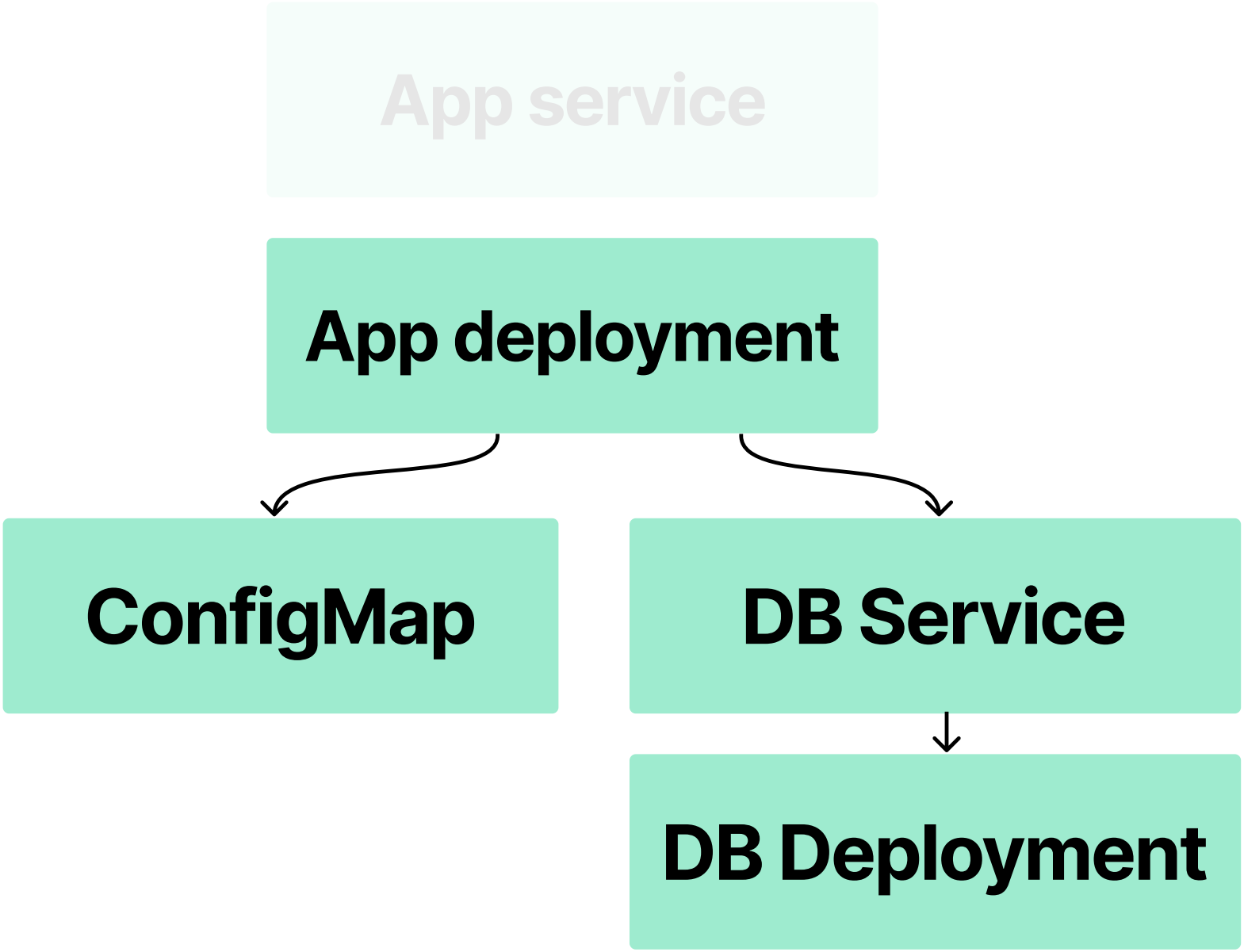
App deployment

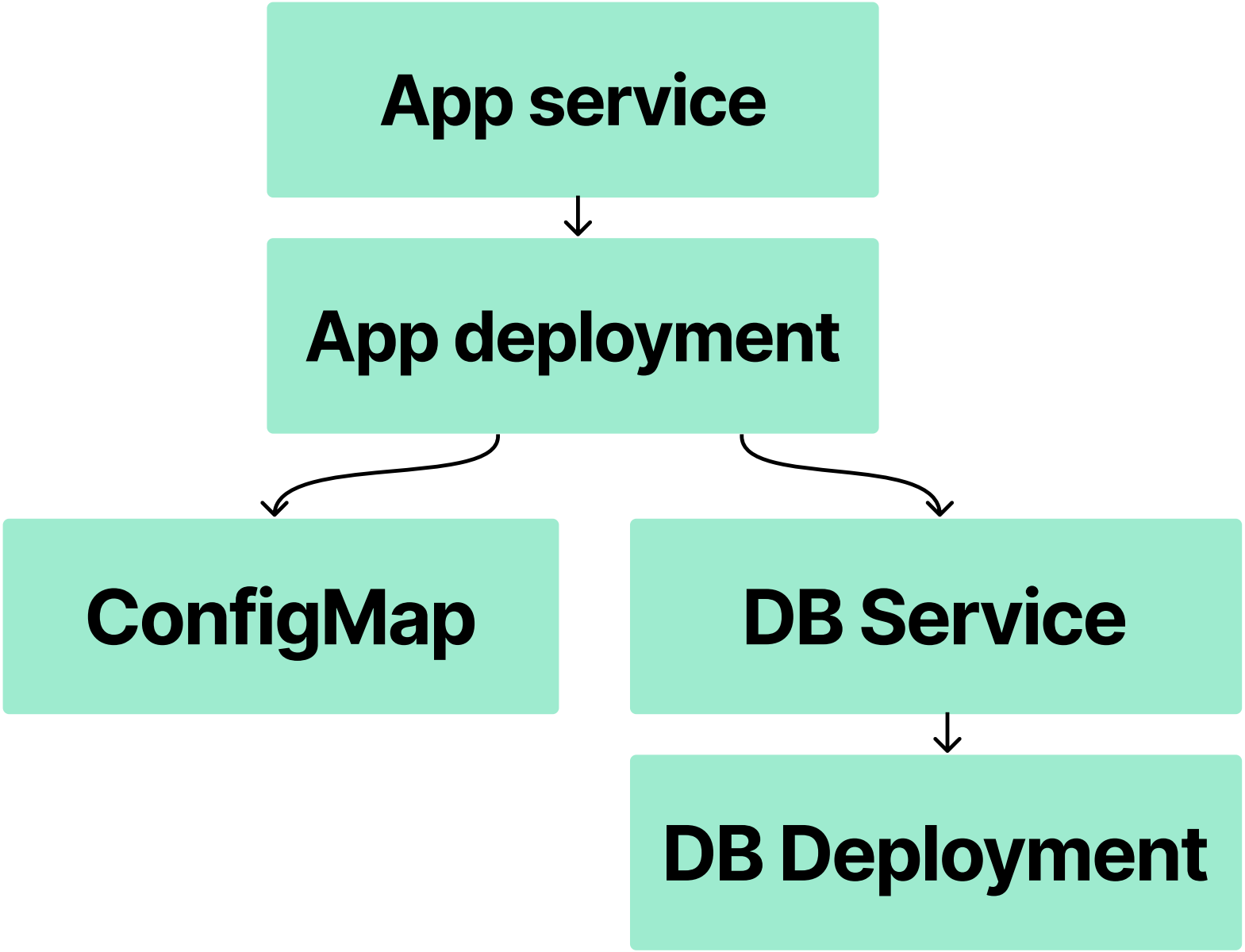
ConfigMap

DB Service

↓

DB Deployment


App service
App deployment
ConfigMap
DB Service
DB Deployment


App service
App deployment
ConfigMap
DB Service
DB Deployment

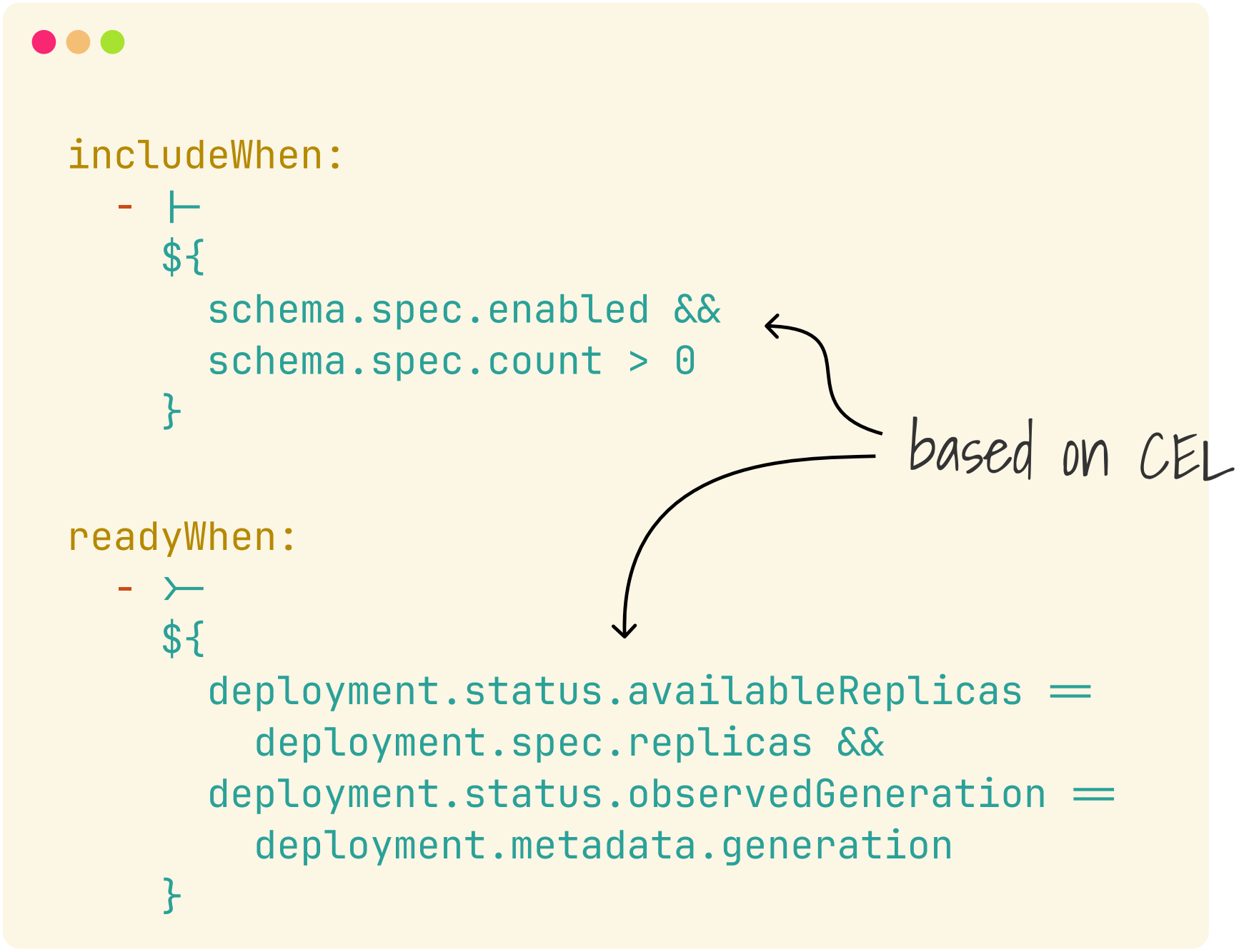
Conditions and expressions



```
includeWhen:
  - ├─
    ${
      schema.spec.enabled &&
      schema.spec.count > 0
    }
```

```
includeWhen:
  - ├─
    ${
      schema.spec.enabled &&
      schema.spec.count > 0
    }

readyWhen:
  - ›─
    ${
      deployment.status.availableReplicas ==
        deployment.spec.replicas &&
      deployment.status.observedGeneration ==
        deployment.metadata.generation
    }
```


includeWhen:
- ├─
${
schema.spec.enabled &&
schema.spec.count > 0
}
based on CEL
readyWhen:
- ›─
${
deployment.status.availableReplicas ==
deployment.spec.replicas &&
deployment.status.observedGeneration ==
deployment.metadata.generation
}

```yaml
# Merge user labels with default labels
labels: ${{"app": schema.spec.name, "managed-by": "kro"}.merge(schema.spec.extraLabels)}

# Parse a JSON string from a ConfigMap or user input
config: ${json.unmarshal(schema.spec.jsonConfig)}

# Printf-style formatting
roleArn: ${"arn:aws:iam::%s:role/%s".format([schema.spec.accountId, schema.spec.roleName])}

# Slice a list
subset: ${schema.spec.items.slice(0, 3)}

# Encode a hash to base64
encoded: ${base64.encode(hash.sha256(schema.spec.data))}

# Extract the host from an endpoint URL
host: ${url(schema.spec.endpoint).getHostname()}

# Extract version numbers from a string
versions: ${"v1.2.3 and v4.5.6".findAll('[0-9]+\\.[0-9]+\\.[0-9]+')}

# Arithmetic on quantities
total: ${quantity(schema.spec.cpuRequest).add(quantity('100m'))}
remaining: ${quantity('2Gi').sub(quantity(schema.spec.memoryRequest))}

# Sum all replica counts
totalReplicas: ${schema.spec.replicaCounts.sum()}

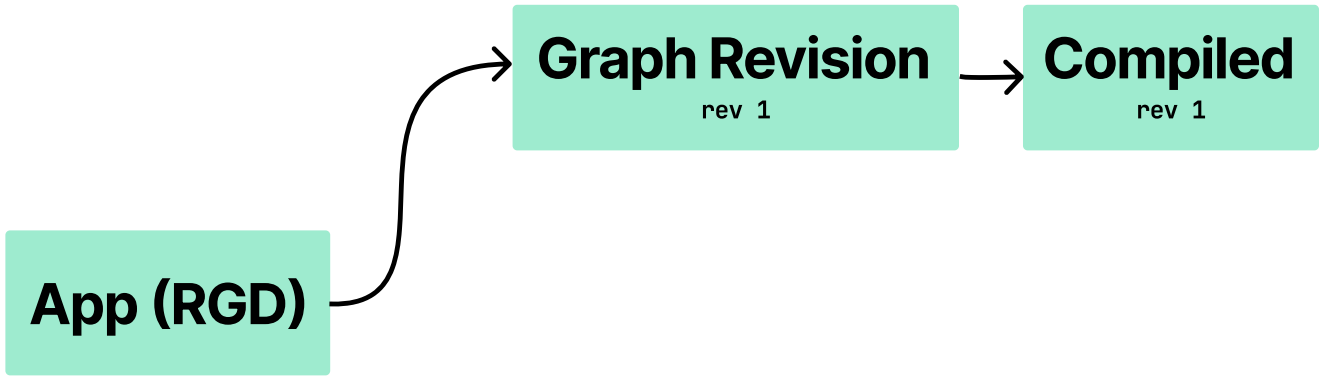
# Validate an IP address from user input
ready: ${isIP(schema.spec.clusterIP)}
```

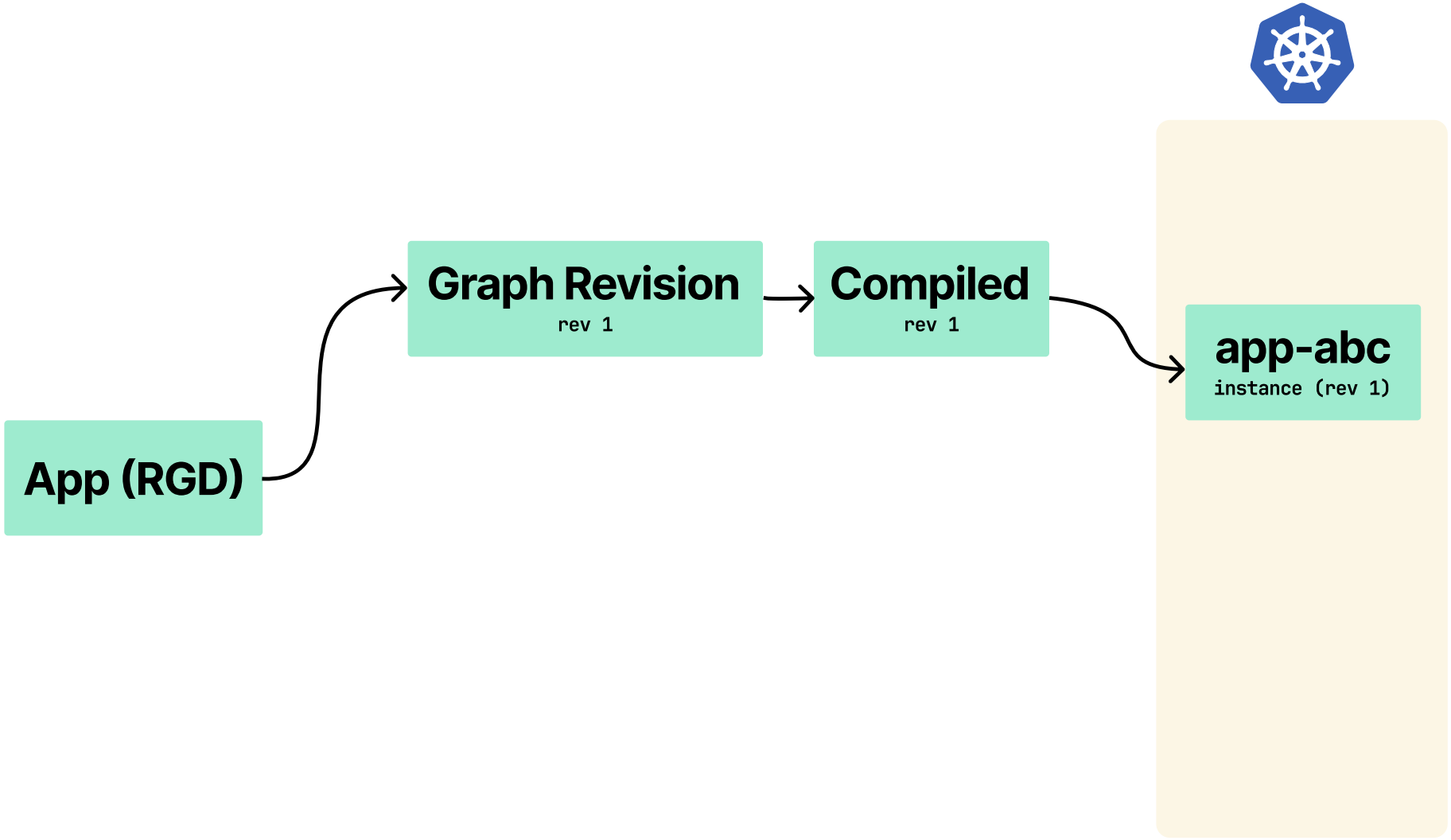
built-in libraries

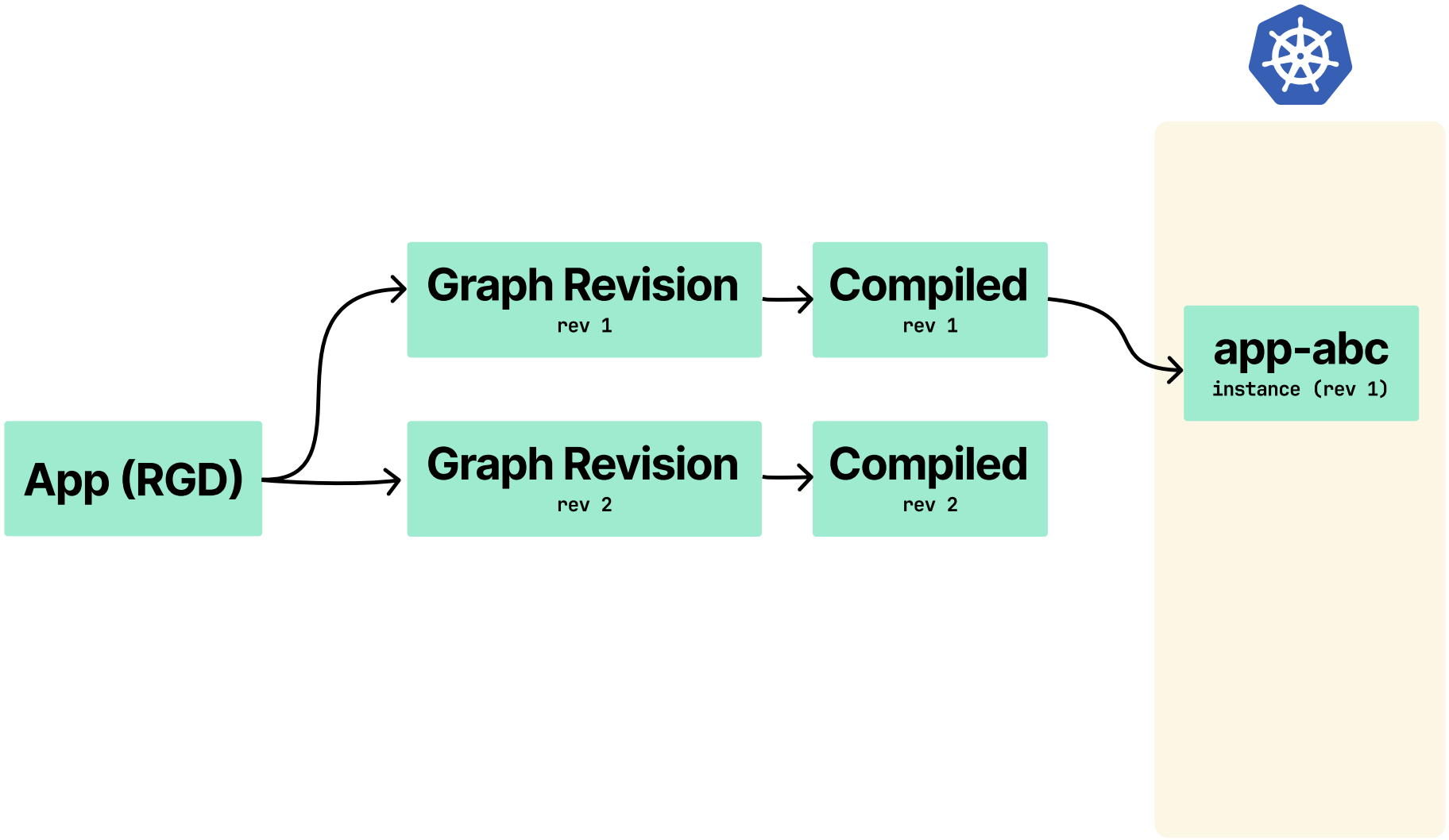
Graph revisions

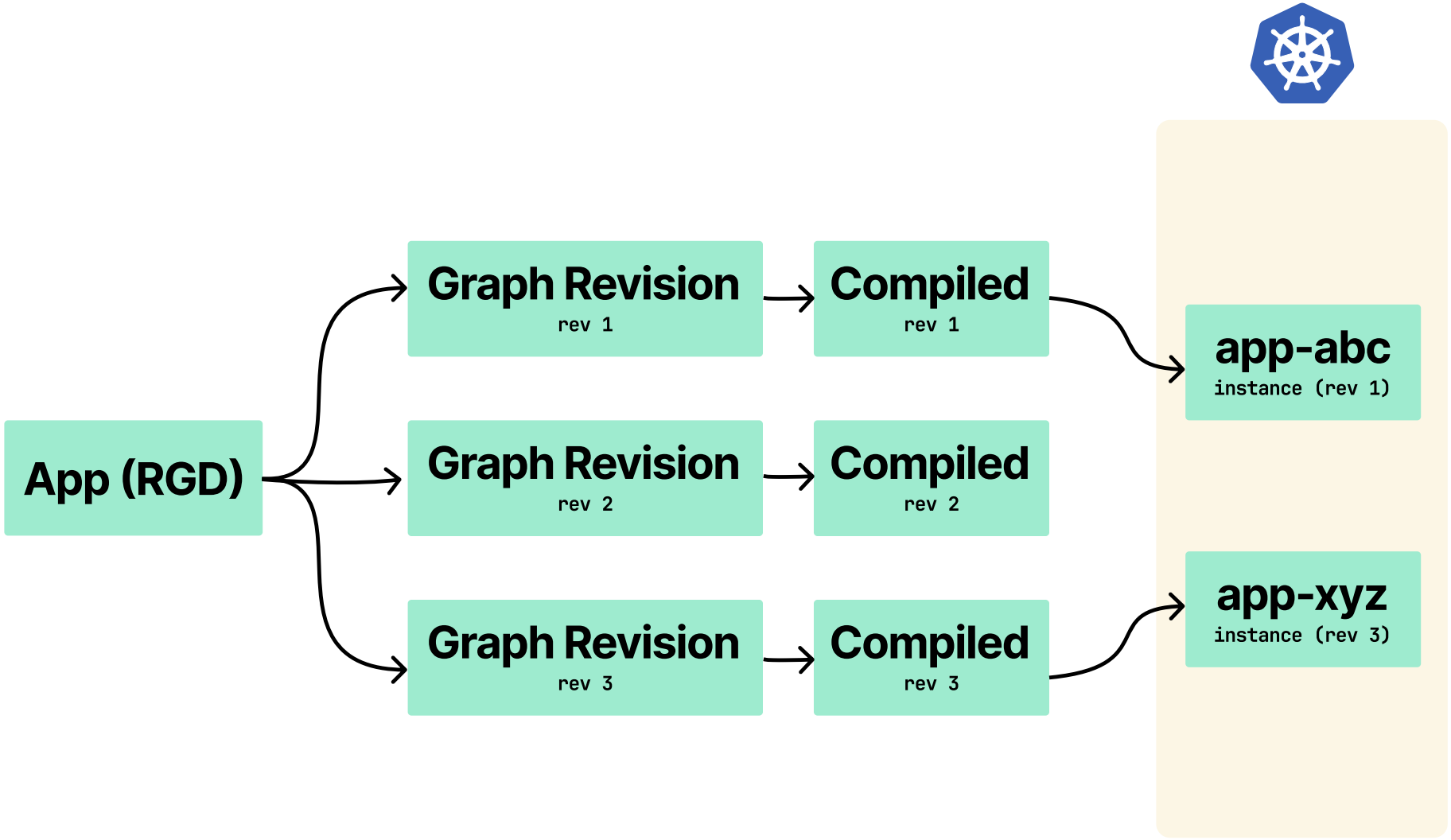


App (RGD)

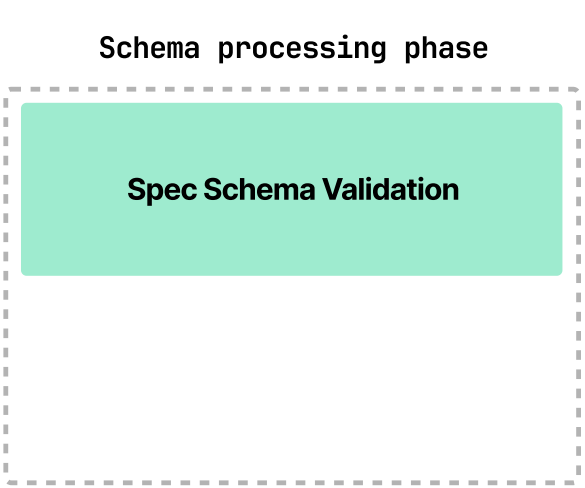

Graph Revision
rev 1
Compiled
rev 1
App (RGD)

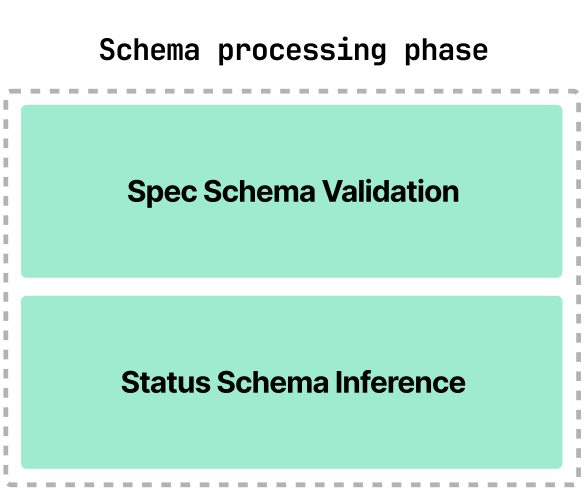

App (RGD)
Graph Revision
rev 1
Compiled
rev 1
app-abc
instance (rev 1)

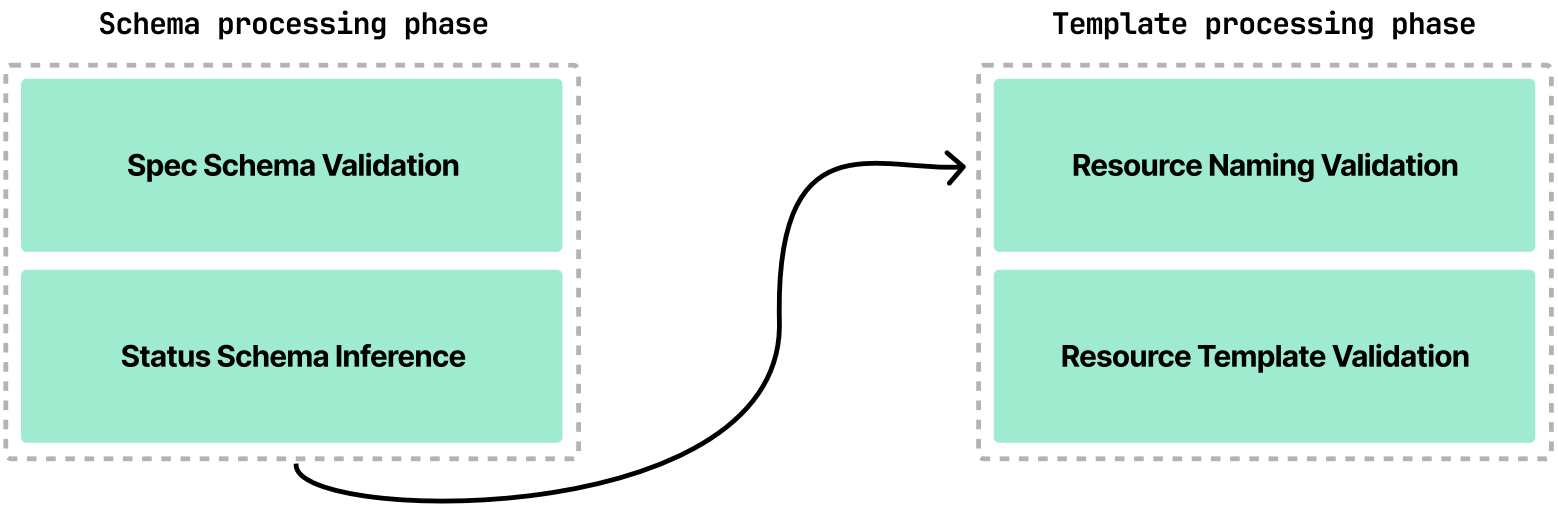

App (RGD)
Graph Revision
rev 1
Graph Revision
rev 2
Compiled
rev 1
Compiled
rev 2
app-abc
instance (rev 1)

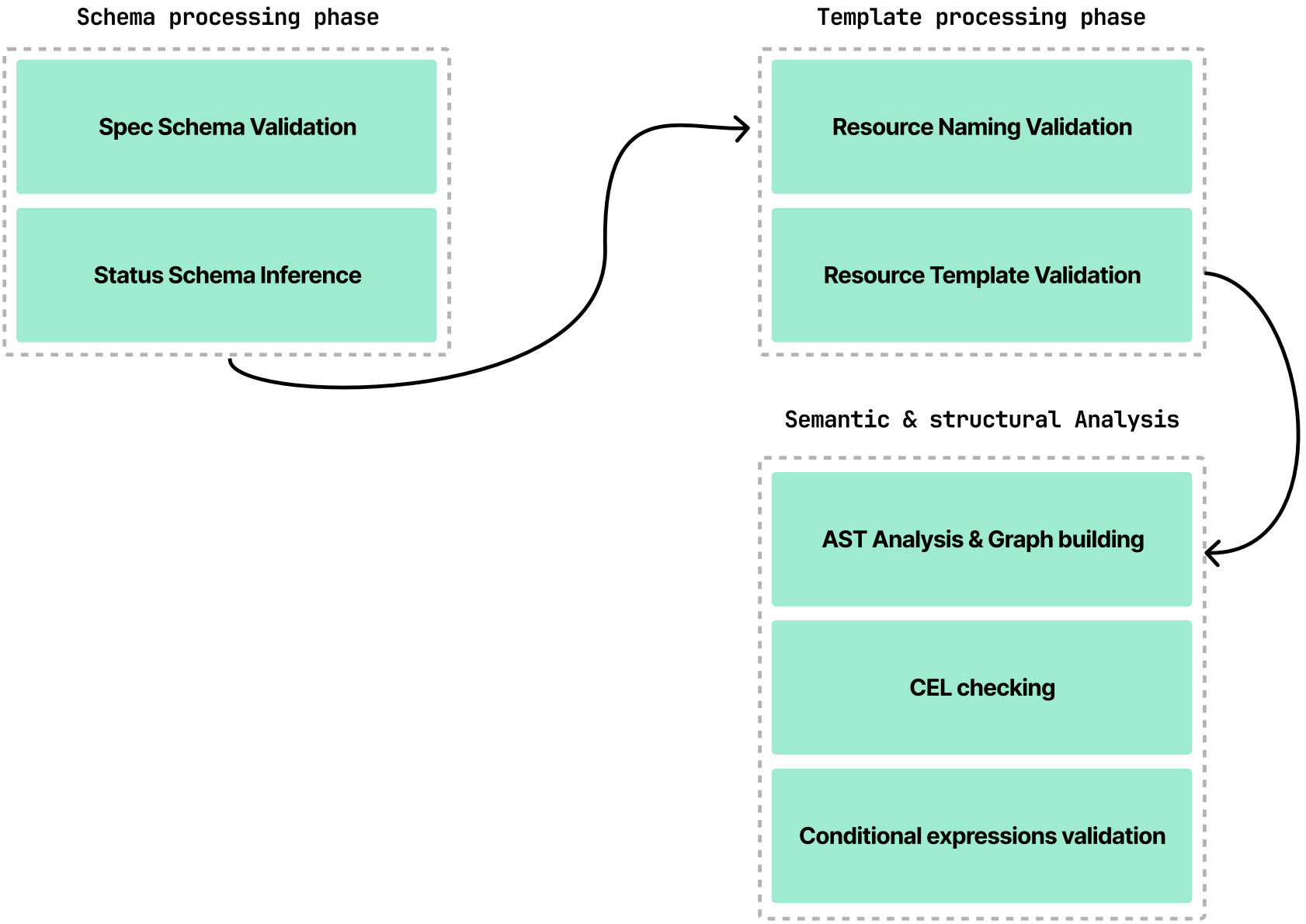

App (RGD)
Graph Revision
rev 1
Graph Revision
rev 2
Graph Revision
rev 3
Compiled
rev 1
Compiled
rev 2
Compiled
rev 3
app-abc
instance (rev 1)
app-xyz
instance (rev 3)

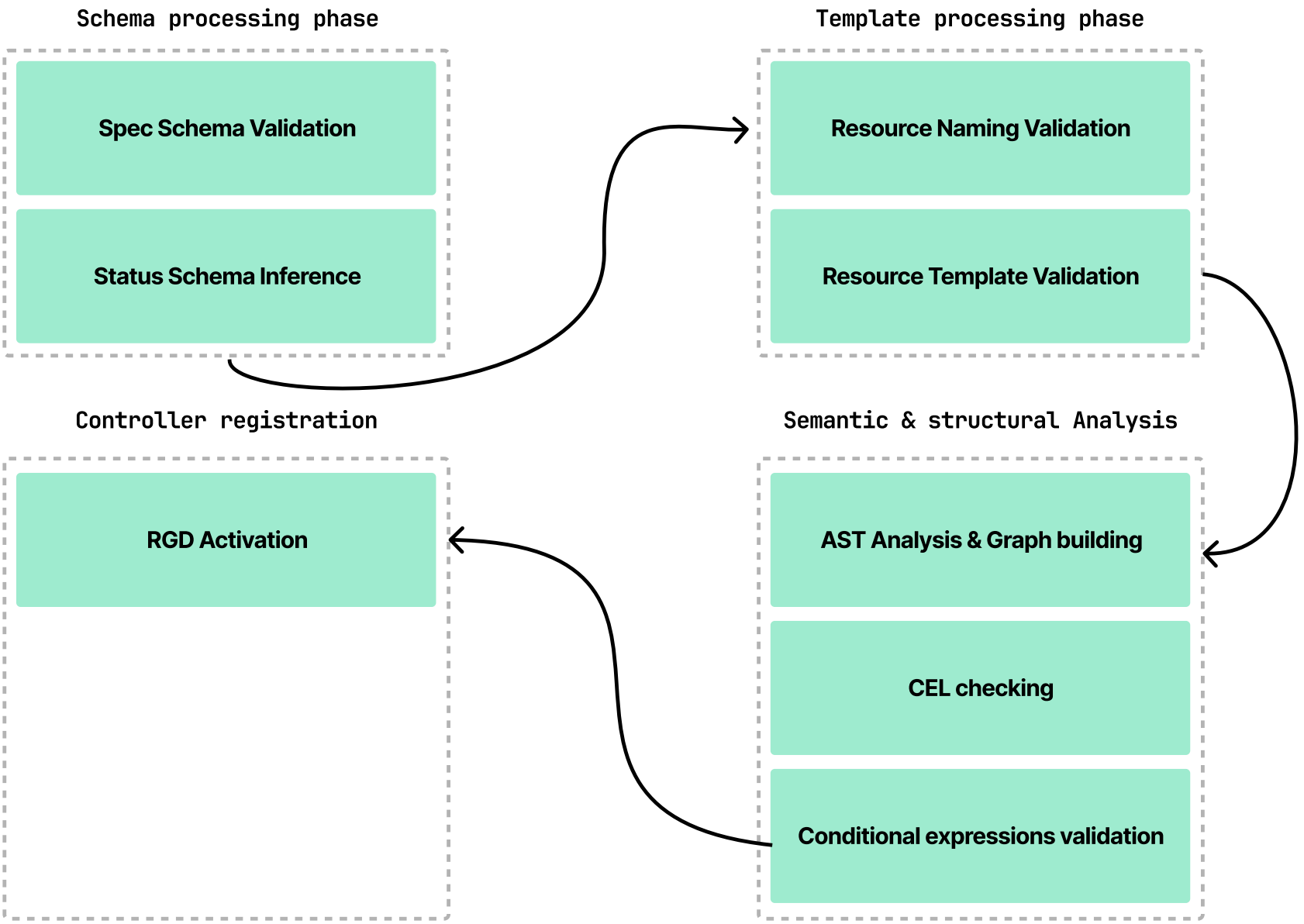
Static analysis


Schema processing phase
Spec Schema Validation


Schema processing phase
Spec Schema Validation
Status Schema Inference


Schema processing phase
Spec Schema Validation
Status Schema Inference
Template processing phase
Resource Naming Validation
Resource Template Validation


Schema processing phase
Spec Schema Validation
Status Schema Inference
Template processing phase
Resource Naming Validation
Resource Template Validation
Semantic & structural Analysis
AST Analysis & Graph building
CEL checking
Conditional expressions validation


Schema processing phase
Spec Schema Validation
Status Schema Inference
Template processing phase
Resource Naming Validation
Resource Template Validation
Controller registration
RGD Activation
Semantic & structural Analysis
AST Analysis & Graph building
CEL checking
Conditional expressions validation

Templating with code

```
~$ cat pod.js
```

```
const { Pod, Container } = require('kubernetes-models/v1')

const pod = new Pod({
  metadata: {
    name: 'test-pod',
  },
  spec: {
    containers: [
      new Container({
        name: 'test-container',
        image: 'nginx',
        env: [{ name: 'ENV', value: 'production' }],
      }),
    ],
  },
})

// Any valid JSON is also valid YAML
const json = JSON.stringify(pod, null, 2)

console.log(json)
```

```
~$
```

```
~$ cat pod.js

const { Pod, Container } = require('kubernetes-models/v1')

const pod = new Pod({
  metadata: {
    name: 'test-pod',
  },
  spec: {
    containers: [
      new Container({
        name: 'test-container',
        image: 'nginx',
        env: [{ name: 'ENV', value: 'production' }],
      }),
    ],
  },
})

// Any valid JSON is also valid YAML
const json = JSON.stringify(pod, null, 2)

console.log(json)

~$
```

```
~$ cat pod.js

const { Pod, Container } = require('kubernetes-models/v1')

const pod = new Pod({
  metadata: {
    name: 'test-pod',
  },
  spec: {
    containers: [
      new Container({
        name: 'test-container',
        image: 'nginx',
        env: [{ name: 'ENV', value: 'production' }],
      }),
    ],
  },
})

// Any valid JSON is also valid YAML
const json = JSON.stringify(pod, null, 2)

console.log(json)

~$
```

Kubernetes SDKs

```
public static void main(String[] args) throws Exception {
  String environment = "production";
  Pod pod = createPod(namespace);

  if (args[0].equals("yaml")) {
      System.out.println(SerializationUtils.dumpAsYaml(pod));
  } else {
      System.out.println(mapper.writeValueAsString(pod));
  }

}
```

```
public static Pod createPod( String environment ) {
    return new PodBuilder().withNewMetadata()
            .withName("test-pod")
            .endMetadata()
            .withNewSpec()
            .addNewContainer()
            .withName("test-container")
            .withImage("k8s.gcr.io/busybox")
            .withEnv(new EnvVarBuilder()
        .withName("ENV").withValue(environment).build())
            .endContainer()
            .endSpec()
            .build();
}
```

```
from kubernetes import client, config

def create_pod(environment):
    return client.V1Pod(
        api_version = "v1",
        kind = "Pod",
        metadata = client.V1ObjectMeta(
            name = "test-pod",
        ),
        spec = client.V1PodSpec(
            containers = [
                client.V1Container(
                    name = "test-container",
                    image = "nginx",
                    env = [
                        client.V1EnvVar(
                            name = "ENV",
                            value = environment,
                        )
                    ]
                )
            ]
        )
    )
```

```
func createPod(environment string) *corev1.Pod {
 return &corev1.Pod{
  TypeMeta: metav1.TypeMeta{
   Kind:       "Pod",
   APIVersion: "v1",
  },
  ObjectMeta: metav1.ObjectMeta{
   Name: "test-pod",
  },
  Spec: corev1.PodSpec{
   Containers: []corev1.Container{
    {
     Name:  "test-container",
     Image: "nginx",
     Env: []corev1.EnvVar{
      {
       Name:  "ENV",
       Value: environment,
      },
     },
    },
   },
  },
 }
}
```

Infrastructure as Code

```
import * as k8s from "@pulumi/kubernetes";

const pod = new k8s.core.v1.Pod("example", {
    spec: {
        restartPolicy: "Never",
        volumes: [
            {name: "shared-data", emptyDir: {}},
        ],
        containers: [
            {
                name: "nginx",
                image: "nginx",
                resources: {requests: {cpu: "50m", memory: "50Mi"}}
            }
        ],
    }
}, { provider: provider });
```

```
const labels = { app: 'guestbook', tier: 'frontend' };

new k8s.Deployment(this, 'deployment', {
  spec: {
    selector: { matchLabels: labels },
    replicas: 3,
    template: {
      metadata: { labels },
      spec: {
        containers: [
          {
            name: 'php-redis',
            image: 'gcr.io/google-samples/gb-frontend:v4',
            ports: [{ containerPort: 80 }],
          }
        ]
      }
    }
  }
});
```

Templating with code (2)

Templating with code (2)

Familiar language

Interpolation, fn

Deeper integration

Templating with code (2)

Familiar language

Interpolation, fn

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Templating with code (2)

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Recap

Recap

1. yq

Recap

1. yq

2. Kustomize

Recap

1. yq
2. Kustomize
3. Helm

Recap

1. yq
2. Kustomize
3. Helm
4. KRO

Recap

1. yq
2. Kustomize
3. Helm
4. KRO
5. **Templating with code**

Thanks.